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In Re: United States v. Anatoly Legkodymov, Criminal Docket No. 23-496

Expert Report

Dr. Boris M. Richard, Ph.D.

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Table of Contents

I. Introduction

UpAI, LLC ("UpAI") has been retained by Venable LLP ("Counsel") and Mr. Anatoly Legkodymov ("Client") to rebut certain claims contained in the United States of America Amended Affidavit and Complaint in Support of an Application For an Arrest Warrant dated January 14, 2023 ("Amended Affidavit and Complaint"), including but not limited to the alleged volume of cryptocurrency transfers between Hydra and Bitzlato wallets. I have also been asked to review and provide my opinion on the extent of efforts undertaken by Bitzlato to develop, deploy and implement at-scale "know-your-customer" and anti-money laundering compliance procedures and protocols.

My name is Dr. Boris M. Richard and I am the Founder and CEO of UpAI, LLC. Prior to starting my consulting company, I served for two years as the Global Head of Crypto Advisory and Digital Asset Economic Disputes at JS Held. Prior to my employment with JS Held, I worked for four years at FTI Consulting where I led the cryptocurrency advisory and digital asset litigation expert services as part of the Forensic and Litigation Consulting division. Earlier at Cornerstone Research, I was engaged for seven years as a litigation consulting expert focusing on high-profile complex disputes involving the valuation and performance analysis for various types of fixed income securities and their derivatives, as well as analyses of institutional trading in various securities markets. My litigation consulting expertise in digital assets and cryptocurrency markets draws upon my six-year crypto advisory experience and more than 15-year long career as a securities strategist, bond trader and portfolio manager at major broker-dealers and large established hedge funds. I graduated from Purdue University with a doctorate in Economics. I am also a native Russian speaker.[1]

My blockchain and cryptocurrency experience includes analysis of the economics and funding of blockchain projects, investment and consumptive uses of native cryptographic assets, cryptocurrency issuance practices, as well as trading of digital assets on centralized and decentralized trading platforms, and analysis of market manipulation practices in cryptocurrencies. I also have extensive expertise in analyzing blockchain network activity and transactional history of specific blockchain addresses and cryptocurrency wallets using commercially available forensic software, as well as publicly available block explorer applications. My experience also includes valuation of digital tokens across various blockchains. I am well familiar with existing industry and scholarly research on various types of permissionless public blockchains, their purpose, functionality, consensus protocols, as well as various kinds of associated cryptocurrencies.

A copy of my current curriculum vitae is attached as Appendix 1 to this Report. For my services related to this matter, I was retained by the defense team and compensated for my time. My compensation is not contingent upon its conclusions or opinions expressed herein, nor upon the outcome of any matter in which these conclusions are presented. While I was able to review certain document and information produced by Client and Bitzlato in this matter, I fully reserve the right to amend or revise the results of the analysis in this report if and when new relevant information becomes available to me.

[1] I was able to review some internal Bitzlato communications and documents in Russian.

II. Summary of Findings

My analysis of the documents produced in this case, my interview with Client and former Bitzlato employees, my examination of the publicly available blockchain data and other information, as well as findings by independent blockchain analytics providers, have allowed me to reach the following conclusions.

First, there is no basis to believe that the DOJ estimated indirect transfers between Hydra and Bitzlato are accurate or reliable. In fact, publicly available information, as well as analyses performed by independent blockchain analytics providers, indicate that such estimates are substantially biased upwards. Furthermore, given the lack of trusted commercial tools at the time, it would have been very difficult, if not impossible, for Bitzlato to detect in real-time indirect transfers between its wallets and Hydra wallets in 2018- 2020.

Second, the degree of exposure of Bitzlato platform to Hydra Market can only be properly assessed when one takes into account the timing of the availability of proven commercial compliance screening tools for addresses and transactions in the crypto industry. My review shows that 60 percent or more of Bitcoin transfers between Hydra and Bitzlato had already taken place before Bitzlato could have access to effective compliance screening capabilities to detect such transfers and to block the appropriate user accounts on its platform.

Third, Bitzlato compliance team worked to build and implement AML compliance protocols consistently in 2021-2022 (the first time robust at-scale real-time compliance monitoring tools became available), but its efforts were materially hampered by technical issues, as well as the refusal of major US blockchain monitoring and compliance companies to provide services to Bitzlato despite the latter's repeated attempts. Notwithstanding all the challenges, Bitzlato compliance related efforts resulted in a significant reduction of illicit cryptocurrency transactions on the platform starting from late 2021.

Fourth, Mr. Legkodymov did not deliberately attempt to avoid implementing KYC/AML policies – in contrast, he pushed his staff for a faster implementation of such policies and was genuinely frustrated by technical setbacks and slow responsiveness of his staff. The record clearly shows that Mr. Legkodymov's efforts were the direct opposite of any conspiracy to launder crypto funds together with Hydra Market, i.e., they involved freezing of accounts linked to Hydra, and therefore a positive blocking of Hydra activities even before law enforcement moved against Hydra. Additionally, Mr. Legkodymov was never accused of violating any anti-money laundering laws and regulations of either Hong Kong or the United States. Even with Mr. Legkodymov's entry into the United States months before his arrest and the years of cooperation between Bitzlato and United States law enforcement, starting from June 2021, there were no indications that he was running afoul of the US law.

Fifth, my review of Bitzlato trading records shows that the overwhelming majority of transactions on the Bitzlato platform was conducted by users from Russia and the republics of the former Soviet Union in currencies other than USD. Being a foreign-registered company with miniscule to no exposure to the United States, Bitzlato was not a money transmitter business (MSB) subject to FinCEN registration requirements and associated AML/CFT obligations under the Bank Secrecy Act, nor was it ever accused of violating such requirements and obligations.

Finally, based on my comparative review of criminal money laundering claims and alleged violations brought by the U.S. Government and regulatory agencies against much larger cryptocurrency exchanges Binance and BitMEX, Bitzlato's exposure to illicit finance was far less than that of these two exchanges. Furthermore, unlike these two exchanges and their criminally charged former CEOs, Bitzlato and Mr. Legkodymov did not seek to deliberately avoid AML/KYC requirements. Unlike Binance and BitMEX's senior executives, Mr. Legkodymov did not engage in a willful violation of the Bank Secrecy Act and its implementing regulations, and his record cannot be any farther from the behavior and motivations of a person who would seek to conspire with a darknet operator to launder illicit funds. In reviewing the conduct of the senior executives at Binance and BitMEX and comparing it to the efforts made by Mr. Legkodymov, it is hard to see similarities. Mr. Legkodymov's internal communications with the Bitzlato team demonstrate his commitment to advancing compliance and overview and focus on cooperation with law enforcement in many countries.

III. There is No Basis to Believe That the DOJ Estimates of Indirect Transfers Between Hydra and Bitzlato Are Accurate or Reliable; In Fact, They Are Significantly Overestimated

Tracing the flow of cryptocurrencies between various pseudo-anonymous entities requires the knowledge of addresses (public keys) of these entities, and address ownership attribution (so-called wallet labeling or tagging) is often not a trivial exercise in itself because blockchain addresses are mostly anonymous. Once the ownership of addresses is established, however, tracking transfers between two addresses no longer presents challenges so long as there are no intermediate addresses between the origin of funds and their destination.

Things become much more complex, however, if transfers are indirect, i.e. have intermediate addresses (wallets) between the origin and destination. In these situations, cryptocurrency moves along more than one step (hop), and it becomes critically important to establish that the ownership of funds does not change along the way. Tracing indirect Bitcoin transfers along multiple hops has an added complexity due to the structure of Bitcoin payment transactions which allows multiple sources of funds (so-called inputs) and multiple destinations of funds (so-called outputs) in one single transaction. A well-defined methodology for estimating indirect transfers, therefore, becomes very important.

This section of the present report discusses challenges of tracing indirect Bitcoin transfers in more detail and lays out arguments why the DOJ estimates of indirect Bitcoin transactions between Hydra and Bitzlato addresses are unreliable and significantly overestimated. I also believe that given the lack of proven commercial tracing tools to assess at-scale and in real-time the risk of illicit flows with intermediate hops in 2018-2019 and for most of 2020, it would be unfair today to allege in retrospect that Bitzlato could have and should have blocked interactions between its users and Hydra wallets when no such reliable tracing tools were available at that time. Bitzlato did, in fact, freeze a significant number of Hydra-linked accounts when it had evidence of such, and it was their policy to do so.

1. Tracing indirect Crypto Flows is More Art Than Science

"For all of 2022, and the brief weeks that Bitzlato operated in 2023, only $9.7 million moved directly from Bitzlato to Binance, according to data from blockchain tracing company Arkham Intelligence. In the four years that Bitzlato operated, only $52 million moved directly from the exchange to Binance, the same dataset shows." "Millions more in smaller transactions ultimately ended up in Binance's wallets. The on-chain data can't account for any additional funds that moved to Binance from Bitzlato through mixers, services that allow users to obfuscate the origin and endpoint of their crypto."[2]

The two quotes above from a 2023 CNBC article highlight the fact that indirect cryptocurrency flows dominate the overall on-chain transaction volumes and that often times it is difficult or impossible to quantify them with a reasonable degree of reliability.

Blacklisting, or tracing illicit (tainted) crypto assets through intermediate addresses, is challenging because the ownership of anonymous wallets may well change along the way, because there is often no exact match between a traced inflow and an outflow, and because networks like Bitcoin allow transactions aggregating multiple sources of funds (both legitimate and illicit) to make transfers to multiple addresses. Such challenges are widely acknowledged by the crypto industry, extensively studied in academic literature, and remain significant even now.

Chart 1 depicts an example of a two-hop transfer of Bitcoin from Hydra to Bitzlato wallet through an unlabeled wallet ~3G2M2. During the first hop, a total of 9 Hydra addresses sent 0.122957 BTC (9 inputs) to 6 receiving addresses (6 outputs), of which 0.0109698 BTC was sent to address ~3G2M2[3]. Chart 1 only depicts the transfer funds from Hydra to the unlabeled intermediary wallet at issue. During the second hop, four addresses belonging to the same wallet as our intermediary address of interest, sent 0.023846 BTC to two receiving addresses from unrelated wallets, of which Bitzlato address ~15SsA received 0.020092 BTC[4]. However, for the purpose of tracing funds flowing from Hydra to Bitzlato, we can only record 0.0109698 BTC, and not the total of 0.020092 BTC received by Bitzlato address during the second hop. This example clearly shows that the prevalence of multi-input/multi-output transactions on Bitcoin network makes it necessary to trace funds at the transaction level, and not at the level of sends and receipts of funds by individual addresses.

2. There Exist Different Methods of Tracking Indirect Bitcoin Flows, and Challenges of Quantifying BTC Flows Are Well Recognized by Both Practitioners and Academics

A 2021 study found "…that a lot of work needs to be done and many challenges need to be addressed for implementing effective AML measures in cryptocurrencies and payment networks"[5]. Although the idea of blacklisting seems simple and intuitively appealing, there are several challenges that might inhibit its practical enforcement. In particular, the taint method used may lead to huge differences in the estimated amount of tainted crypto funds involved in multi-hop payment transactions. The study showed, for

[2] https://www.cnbc.com/2023/01/19/binance-was-final-destination-for-millions-in-funds-from-bitzlato.html.

[3] BTC transaction hash 5b21587ed8216f62cd7a853e1a34283ef7277f47083c8e094f8162a44c5ad4a5.

[4] BTC transaction hash 37ac362b647aaa1e8d234a188c37dda4edffb85442cd2c6dc0845fdea3ee7f2e.

[5] Source, SoK: Money Laundering in Cryptocurrencies, ACMDL Digital Library, Aug 2021 https://dl.acm.org/doi/fullHtml/10.1145/3465481.3465774#BibPLXBIB0002.

example, that a haircut taint method in multi-hop transactions may produce estimated tainted amount 7 times smaller than those estimated using poison taint method.[6]

There are several tainting methods in place which have been created to tackle serious tracking issues caused by the UTXO nature of the Bitcoin network.[7] Specifically, when targeted ("tainted") inputs are combined with other unrelated inputs in the same wallet into new output(s), it is difficult to identify or differentiate the exact distribution and destination of the targeted inputs without having a precise methodology. In our specific case, when Bitcoin is moved from a Hydra/Bitzlato address to a Bitzlato/Hydra address not directly, but through an intermediary address which has multiple previously obtained UTXOs in its balance, it is far from obvious whether the output received by Bitzlato/Hydra address represents the input/amount which had been received by the intermediary wallet from Hydra/Bitzlato.

Poison and Haircut methods classify all of the transaction outputs as tainted outputs in the transactions with any tainted input, and while Haircut method distributes tainted coins to all outputs proportionally, in the presence of mixing clean and tainted Bitcoin, the tainting result of both methods accumulates an excessive number of tainted transactions and addresses. The two tracking methods simply overestimate the number of "tainted" transactions and the amount of "tainted" BTC transferred. While more practical for tracking purposes, alternative tracking methods like FIFO (first-in/first-out) and LIFO (last-in/first-out) methods likewise may not produce accurate tracking results for transactions with multiple inputs because the distribution of amounts of transferred BTC across outputs is arbitrary and tracked Bitcoin may well end up directed to incorrect addresses.[8]

Academic research showed that various tracking methods discussed above in themselves have quite low accuracy. First, these methods in general are able to trace only 40-60% of stolen BTC to the addresses of known crypto exchanges or known mixers. [9] At the same time, these mechanical methods significantly overestimate the number of "tainted" BTC transactions versus their true number obtained when address ownership is added to the information set – by anywhere from 10% to 90%.[10] In summary, if the DOJ used any of these tracking\tainting methods to estimate the volume of multi-hop transactions between Hydra and Bitzlato wallets, such alleged volumes are likely to be unreliable.

3. MIT Research Study Highlights Challenges in Detecting and Preventing Indirect Illicit Crypto Transfers, and Implies that Indirect Transfers of BTC Between U.S. KYC/AML Compliant Exchange Coinbase and Hydra Market Could be Almost 30 Times Greater Than All Transactions Between Hydra and Bitzlato

The inherent difficulty in accurately quantifying flows of Bitcoin from one entity to another through intermediary addresses in the form of multi-hop transfers was noted by the MIT and London School of

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[6] *Id*., Table 1.

[7] Bitcoin balance in each address exists in the form of unspent transaction outputs (UTXO), which are newly created from the sum of inputs in that transaction.

[8] Tin Tironsakkul *et al*., *Context Matters: Methods for Bitcoin Tracking*, Forensic Science International: Digital Investigation, October 2022. Authors note than another method, Taint-in/highest-out (TIHO), does not perform well when the flow of Bitcoin does not involve mixers along the way.

[9] *Id*., Figure 5, Panels (a), (b) and (c).

[10] *Id*., Figure 7.

Economics researchers in the context of the efficacy of KYC/AML policies.[11] As part of their analysis of the transactional history and on-chain interactions of Hydra Market, the researchers estimated that while Coinbase, a major US crypto exchange, directly sent and received modest 196 and 126 bitcoins from Hydra Market, respectively, it sent 530,000 and received 218,000 bitcoins via the neighboring clusters of addresses between 2020 and June 2021[12]. These numbers are almost thirty times higher than 25,465 BTC estimated by Arkham Intelligence to have been transferred between Hydra and Bitzlato between May 2018 and April 2022, and dwarf approximately 142,500 BTC transacted by ALL counterparties with Bitzlato during a much longer period between December 2015 and November 2022.[13] Indeed, it is very challenging to trace the origin of Bitcoin and quantify its flow when indirect transactions involve intermediary addresses to obfuscate the source of funds, and there is absolutely no basis to believe that a peer-to-peer trading platform like Bitzlato would have better means of detecting and blocking indirect transactions with Hydra than a well-capitalized US cryptocurrency exchange which holds BitLicense in New York state and is universally regarded as the most regulated and compliant crypto trading platform in the United States and globally. Tellingly, the authors of the research stated that "…even if KYC entities were restricted to deal exclusively with other KYC entities, preventing inflows of tainted funds would still be nearly impossible, unless one was willing to put severe restrictions on who can transact with whom and make every transaction subject to the approval of a blockchain "monitoring entity".[14] But this is simply not how the Bitcoin network works. And yet despite the immensely greater possible level of Hydra-linked transactions into the US-based Coinbase, I am aware of no prosecution of any Coinbase personnel, and certainly not its founders, board members or shareholders.

4. Blockchain Intelligence Company Chainalysis - Frequently Employed by the Government - Recently Admitted Under Oath That There is No Scientific Evidence of the Accuracy of its Cryptocurrency Tracing Software

The crypto industry, exchanges and blockchain analytics firms themselves are well aware of the complexities and common inaccuracies inherent in cryptocurrency tracing, especially when it involves indirect transfers. According to public media reporting, Chainalysis' head of Government Solutions testified in June of 2023 in the criminal case brought by the US Department of Justice against Roman Sterlingov that she was "unaware" of scientific evidence for the accuracy of Chainalysis' investigative software product Reactor. The executive said she was unable to provide the court with statistical error rates for Chainalysis' Reactor software, unaware of any scientific peer-reviewed papers or "anything published anywhere" attesting to the accuracy of Chainalysis Reactor, and testified that Chainalysis reportedly judged its software's accuracy using customer feedback.[15] Such revelations are significant in light of the fact that Chainalysis is one of the leading partners of law enforcement and other government agencies in investigating illicit cryptocurrency transactions and crypto crime and a substantial portion of the Chainalysis revenue is derived from its work with law enforcement in the United States.

[11] Igor Makarov and Antoinette Schoar, *Blockchain Analysis of the Bitcoin Market*, April 18, 2022, London School of Economics and MIT.
[12] *Id*., Figure 8.
[13] Source: Crystal Intelligence data.
[14] *Id*. 4, p. 3.
[15] https://www.coindesk.com/consensus-magazine/2023/07/24/chainalysis-investigations-lead-is-unaware-of-scientific-evidence-the-surveillance-software-works/.

5. Multi-Hop Crypto Funds Tracing Remains a Challenge Up Until Today

The leading and compliant US-based crypto exchange Coinbase published a series of materials in March 2022 on blockchain tracing and stated that "…it's more fitting to consider blockchain analytics more of an art than science" and "…it's fairly evident that an external observer cannot possibly gain a full picture or claim 100% confidence in [wallet] attribution".[16] It also showed that blockchain analytics and wallet screening quickly become challenging at scale as addresses clustered through the commonly accepted "commonspend" heuristic exponentially magnify in terms of their number.[17] "Commonspend" heuristic is the rule by which addresses providing multiple input funds for a given Bitcoin payment transaction must be owned by the same entity because such transactions require shared access to private keys. Private keys, on the other hand, are almost never shared among different entities. For reference, when one uses "commonspend" heuristic alone, the number of Bitzlato addresses exceeds 250,000, while the number of Hydra addresses is in the vicinity of 6.6 million.[18]

A very recent, May 2024, analysis by the blockchain analytics and compliance solution provider TRM Labs highlighted several key considerations for evaluating the risk and amount of indirect transfers on the blockchain.[19] Among other things, TRM argued that there is no single factor determinative to establish the validity of indirect risk and stated that "blockchain intelligence companies like TRM may each have their own methodology on how indirect risk is calculated and presented to users". The company pointed to such factors as number of hops in a chain of transactions, presence of centralized services such as exchanges, payment processors etc., time between hops, additional blockchain data about types of wallets, as well as activity and characteristics of intermediate addresses. On the latter, the prevalence of intermediate addresses with multiple deposit and withdrawal transactions makes a change of ownership more likely, thereby making tracing validity and estimated indirect transfers less reliable.

In short, there is no common proven and reliable method for tracing multi-hop crypto token movements even today, but the Government still contends that Bitzlato and Legkodymov should be liable for movements into or out of their exchange if the Hydra link occurred over multiple hops (sometimes 5 or more).

6. Automated Trusted and At-Scale Multi-Hop Screening Tools Were Largely Unavailable in the Private Crypto Industry in 2018-2020

Complexities inherent in tracing and quantifying indirect crypto flows explain well the lack of trusted tools enabling virtual asset service providers to detect multi-hop illicit transfers in 2018-2019 and most of 2020. Chainalysis introduced automated peel chain detection in November 2020 which captures only one of multiple patterns of indirect transfers for illicit purposes. Another tracing software provider Merkle Science released 5-hop automated tracing capability at the end of June 2021.[20] Elliptic Investigator - a software to calculate the risk exposure of crypto addresses and wallets at scale, as well as to map out

[16] https://www.coinbase.com/blog/part-1-blockchain-analytics-is-more-of-an-art-than-science.
[17] https://www.coinbase.com/blog/part-2-blockchain-analytics-is-tricky-at-scale.
[18] See, for example, https://www.walletexplorer.com/.
[19] https://www.trmlabs.com/post/key-considerations-for-evaluating-indirect-risk-on-the-blockchain.
[20] https://blog.merklescience.com/technology/why-does-the-crypto-industry-need-to-understand-indirect-risk-exposure.

multi-hop connections between multiple entities – was released in May 2022.[21] CipherTrace Inspector for financial investigations and compliance due diligence with embedded risk scoring of transactions and entities and ability to trace indirect transfers became available in April 2022.[22]

In summary, given that there are inherent difficulties and substantively ad-hoc assumptions involved in multi-hop crypto asset tracing, that different companies use different proprietary methods, that various BTC tainting methods notoriously produce large differences in results, and that the US Department of Justice refused to provide any information or data behind its estimates of indirect transfers, there is no basis to believe that the DOJ estimated indirect transfers of Bitcoin between Hydra and Bitzlato wallets are reasonable or reliable. More importantly, given the lack of proven commercial tracing tools to assess at-scale and in real-time the risk of illicit flows with intermediate hops in 2018-2019 and for most of 2020, it is unfair today, with the advantage of better tools and better wallet attribution, to allege in retrospect that Bitzlato could have and should have blocked interactions between its users and Hydra wallets more than it did.

7. The DOJ Provided No Basis to Believe That Its Indirect Hydra-Bitzlato Transfer Volume Estimates Are Reliable

The DOJ Amended Affidavit and Complaint claims that Hydra users sent $218.7 million to Bitzlato wallets indirectly, i.e. through intermediate non-Bitzlato addresses. The complaint further alleges that Hydra wallets received an additional $191.9 million from Bitzlato wallets indirectly, i.e. through non-Bitzlato intermediate wallets. The same estimated volume of indirect transfers between Hydra and Bitzlato customer wallets is recited in the Sentencing Memorandum.[23]

Despite all the complexities associated with multi-hop tracing of Bitcoin flows using blockchain analytics discussed above, the US Government did not provide any details as to how it has arrived at its alleged estimates of indirect cryptocurrency transfers between Hydra and Bitzlato platforms. First, it did not specify which cryptocurrencies were examined. Second, it did not define over how many hops indirect flows were followed. Third, it did not disclose whether labeled intermediate addresses were included into calculations and how the analysis made sure that intermediate addresses did not change ownership to be included into the calculated totals. Fourth, the Government did not explain which cluster(s) of Bitzlato and Hydra addresses it examined and how it arrived at attributing addresses to either Hydra or Bitzlato. Fifth, the government did not specify whether it followed address-level tracing or transaction-level tracing of UTXO-based cryptocurrencies (such as Bitcoin), while address-level tracing may well produce overestimated numbers. Lastly, if transaction-level approach was used for UTXO cryptocurrencies like Bitcoin, the US Government did not provide any explanation on how it performed matching inputs and outputs in transactions with multiple inputs and outputs which are typical for currencies like Bitcoin.[24]

[21] https://www.elliptic.co/media-center/elliptic-redefines-cryptoasset-forensics-with-one-click-blockchain-investigations-across-thousands-of-assets.

[22] https://ciphertrace.com/wp-content/uploads/2022/04/CipherTrace-Inspector-Financial-Investigations-and-Blockchain-Forensics.pdf.

[23] Sentencing Memorandum, Section I, "Introduction".

[24] My understanding is that Mr. Legkodymov's counsel has requested additional documentation from the government and those requests to provide back up for their analysis have been refused.

The basis for the Government claims that Bitzlato wallets received $15 million in ransomware-related cryptocurrency is also far from clear.[25] On the one hand, the Amended Affidavit and Complaint alleges that Bitzlato received more than $15 million worth of cryptocurrency representing the proceeds of ransomware attacks, "based on blockchain analysis by the FBI and the FBI's own information about the addresses to which ransoms have been paid".[26] On the other hand, the Sentencing Memorandum states that "[r]ansomware actors under investigation by the FBI have funneled more than $1.2 million worth of cryptocurrency through accounts on Bitzlato" and that $15 million estimate was produced by a "commercial blockchain analysis firm". Similar to the Government alleged volume of transfers between Hydra and Bitzlato wallets, I have not seen any information from the Government produced in this matter which would have explained even the bare bones of the analysis performed, i.e. direct vs indirect flows, time period, sources of and/or methods for wallet ownership attribution, transactional data, the name of the commercial blockchain analysis company, etc.

In sum, there is no evidence produced in this matter and available for my review that could indicate that the estimates of indirect transfer volume between Hydra and Bitzlato wallets claimed by the US Government are reliable. The Government has also provided a contradictory story as to how it estimated ransomware-related cryptocurrency proceeds allegedly received by Bitzlato customer wallets.

8. DOJ's Hydra-Bitzlato Transfer Volume Estimates Are Much Higher Than Chainalysis Public Findings

In fact, publicly available information from Chainalysis – a blockchain analytics company well known to provide trusted assistance to the US Government agencies – indicates that transfers of cryptocurrency from Hydra to Bitzlato customers claimed in the Amended Affidavit and Complaint are very significantly overestimated. Specifically, in blogpost dated January 18, 2023, Chainalysis estimated that 26% of $2.5bn in cryptocurrency received by Bitzlato customers in 2019 -2023 came from illicit sources, of which 38.2% came from darknet markets.[27] This implies that Bitzlato customers received approximately $248.3 million from darknet markets in 2019-2023. Moreover, the chart "Top 20 illicit entities sending funds to Bitzlato, 2019-2023" from the same blog post indicates that Hydra specifically sent approximately $220 million to Bitzlato customers during the same period of time. The immediate conclusion is that $389.3 million claimed by the DOJ to be received by Bitzlato addresses from Hydra wallets in 2018-2022 is overestimated by approximately $164 million, or 73% (reasonably assuming that Chainalysis would have estimated $225 million of transfers from Hydra to Bitzlato in the 2018-2022 period giving effect to approximately $5.4 million sent to Bitzlato from Hydra in 2018).

Earlier, in Chainalysis Crime Report 2022, the same company reported that Bitzlato customers received from Hydra approximately $206 million in cryptofunds in 2019-2021.[28] Adding approximately $5 million in 2018 and approximately $20 million in 2022 (estimated by Crystal blockchain tracing company), produces approximately $231 million for the period 2018 – 2022. Consistent with the analysis above, the

[25] Sentencing Memorandum, p.2 and Footnote 2.
[26] Amended Affidavit and Complaint, Para 15.
[27] https://www.chainalysis.com/blog/us-authorities-shut-down-high-risk-exchange-bitzlato/.
[28] Chainalysis, the 2022 Crypto Crime Report, February 2022.

overestimation of the transfers from Hydra to Bitzlato by the US Government analysis is again substantial – by approximately 69%.

Equally interestingly, the US Government claims that, based on FBI knowledge and analysis, Bitzlato customers also received, directly or indirectly, more than $15 million of cryptocurrency representing the proceeds of ransomware attacks.[29] Chainalysis, on the other hand, calculates much lower amount of ransomware fund receipts by Bitzlato customers - $10.4 million in 2019-2023 period (1.6% of illicit receipts)[30] and $9 million in the 2019 – 2021 time period.[31]

9. DOJ's Indirect Hydra-Bitzlato Transfer Volume Estimates Are Much Higher Than Estimates of Other Blockchain Analytics Service Providers as Well

Analyses performed at my request by two other blockchain analytics and compliance companies, Hoptrail[32] and Crystal[33], for the period from May 2018 to April 2022 confirm that the US Government's estimates of indirect transfers of cryptocurrency between Hydra wallets and Bitzlato customer wallets, alleged both in the Amended Affidavit and Complaint and the Sentencing Memorandum, are seriously incorrect to the upside.

Multi-hop cryptocurrency transfers are commonly used to obfuscate the origin of funds, wallet ownership and to complicate tracing the flow of crypto assets. Since Bitzlato was mostly operating as a peer-to-peer platform as opposed to a regulated centralized cryptocurrency exchange, there would be a relatively little need for a Hydra user to engage in a complicated multi-hop transactions to send funds to or receive funds from a Bitzlato counterparty. Nevertheless, I have engaged two independent blockchain analytics and crypto compliance companies, Hoptrail and Crystal, to estimate indirect flows of Bitcoin between Hydra and Bitzlato wallets, with number of transaction hops between 2 (Hoptrail) and 5 (Crystal). Figure 1 reports the results and compares them with the DOJ claims.

As Figure 1 demonstrates, while direct crypto transfer volume estimates largely agree between the US Government and the two private companies and are within 0.4% - 2.7% of each other, the indirect transaction volume estimated by the US Government is several times higher. Specifically, the indirect transfers claimed by the US Government to equal to $410.6 million are 3.75X of $109.4 million estimated by Hoptrail across 2-hop transactions, and 2.7X higher than $153.6 million estimated by Crystal across 5-hop transactions.

Notably, overall transfers of BTC from Hydra to Bitzlato estimated by Hoptrail and Crystal to be in the $192 - $207 million range are much closer to approximately $220 million estimated by Chainalysis for the 2019-2023 time period, than they are to the $389.2 million alleged in the Amended Affidavit and Complaint.[34]

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[29] The Amended Affidavit and Complaint, p. 7.
[30] *Id*., 23.
[31] *Id*., 24, Table "Analysis of a selection of Moscow City cryptocurrency businesses facilitating money-laundering | 2019–2021."
[32] https://www.hoptrail.io/.
[33] https://crystalintelligence.com/.
[34] https://www.chainalysis.com/blog/us-authorities-shut-down-high-risk-exchange-bitzlato/.

It is simply hard to corroborate the Government's calculations. They are at odds with the calculations made by other blockchain analytics and tracing companies.

IV. Effective At-Scale Transaction Monitoring and Compliance Tools with Multi-Hop Tracing Capabilities Were Largely Unavailable for the Private Crypto Industry in 2018 - 2020

It is much easier to estimate the volume of past non-compliant transactions using a large database of labeled blacklisted addresses and multi-hop tracing tools in 2023 than it would be three to five years ago, when many real-time detection methods were unavailable, as was the unfair advantage of a perfect hindsight available today. Bitzlato's record of detecting the interactions of its customers with "dirty" Bitcoin addresses should take into account the availability of effective at-scale address and transaction screening capabilities during relevant times. When technical feasibility (or rather lack thereof) of screening and blocking illicit transactions in a real-time format is put into a proper historical context, it becomes much harder to perceive Mr. Legkodymov and Bitzlato as "bad actors", purposedly enabling illicit transactions with Hydra Market or ransomware-related wallets for profit.

Chainalysis released its platform to perform real-time monitoring of transactions for multiple cryptocurrencies for compliance purposes (Chainalysis KYT) in April 2019[35], while actionable alerts about suspicious transactions did not become available until August 2019.[36] However, Chainalysis' real-time suspicious transaction detection software for the private crypto industry became available only much later, and my examination of public media releases indicates that this process took place in earnest only between March 2021 and April 2022.[37]

The review of product release and partnership information from the second largest blockchain monitoring company Elliptic also indicates that automated and fully integrated AML screening tools became available for private VASPs only in 2021. Actionable information with respect to BTC wallet labels and transaction screening to combat money laundering came from Elliptic in August 2019 when, in collaboration with MIT-IBM Watson AI Lab, it released Elliptic Dataset containing a limited set of 200,000 Bitcoin transactions.[38] Selected crypto industry players partnered with Elliptic in 2020 to monitor transactions, and the firm released its wallet screening tool Elliptic Lens in March 2020.[39] [40] However, the fully automated version

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[35] https://www.prnewswire.com/news-releases/chainalysis-expands-cryptocurrency-support-ahead-of-new-regulatory-guidance-300836844.html?tc=eml_cleartime.

[36] https://www.chainalysis.com/blog/real-time-alerts-press-release/.

[37] https://www.chainalysis.com/blog/chainalysis-bny-mellon-announcement/, https://www.chainalysis.com/blog/chainalysis-capital-union-bank/, https://www.chainalysis.com/blog/chainalysis-cross-river/, https://www.chainalysis.com/blog/robinhood-partnership-announcement/, https://www.chainalysis.com/blog/chainalysis-nydig-bitcoin-balance-sheet/, https://www.chainalysis.com/blog/chainalysis-dapper-labs-announcement/, https://www.chainalysis.com/blog/chainalysis-notabene-travel-rule-integration/.

[38] https://www.elliptic.co/media-center/elliptic-releases-bitcoin-transactions-data

[39] https://www.elliptic.co/media-center/bitcoin-suisse-selects-elliptic-for-crypto-compliance-monitoring, https://www.elliptic.co/media-center/lcx-partners-with-elliptic-to-strengthen-its-crypto-compliance.

[40] https://www.elliptic.co/blog/lens-wallet-screening.

of the blockchain analysis and transaction monitoring tool Elliptic Navigator, which also facilitated the process of SAR submissions, was released only in October 2021.[41] History of partnership engagements between Elliptic and the digital asset industry also confirms that large segments of the crypto industry deployed robust compliance solutions for the first time in 2021.[42]

CipherTrace, first funded by venture capital in February 2019[43], revealed its product Crypto Risk Intelligence in December 2019.[44] Real-time anti-money laundering transaction screening solution for crypto native businesses with predictive risk scoring for transactions CipherTrace Sentry was developed by May 2020[45] and first released in July 2020.[46] CipherTrace Traveler – the first commercial product created to help VASPs to comply with global "travel rule" in digital assets was first released in March 2021[47] and deployed in 2021 as well.[48] CipherTrace Inspector for financial investigations and compliance due diligence with embedded risk scoring of transactions and entities and multihop tracing capabilities became available in April 2022.[49]

My review of public documents also shows that one of the leading providers of blockchain analytics and compliance solutions for the virtual asset service providers, TRM Labs, received its first equity funding round of $1.7mm at the very end of January 2019[50], followed by Round A capital raise of $4.2MM in November 2019.[51] The first compliance monitoring partnership for TRM Labs with crypto trading platform FalconX came only in June 2020.[52] However, TRM crypto compliance and tracing solutions became broadly available to the crypto industry only more than a year later, specifically in September – November 2021, when the firm formed strategic partnerships with FTX US, Solana, Circle Financial, Binance Custody, Anchorage Digital Bank, etc.[53]

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[41] https://www.elliptic.co/media-center/complyadvantage-partners-with-elliptic-for-blockchain-analysis-and-transaction-monitoring.

[42] https://www.elliptic.co/media-center/notabene-and-elliptic-partner-to-provide-a-fatf-compliant-solution, https://www.elliptic.co/media-center/revolut-adopts-elliptics-crypto-compliance-software, https://www.elliptic.co/media-center/hkvax-partners-with-elliptic-to-mitigate-financial-crime-risk.

[43] https://www.securityweek.com/blockchain-security-startup-ciphertrace-emerges-15m-funding/.

[44] https://www.businesswire.com/news/home/20191216005584/en/CipherTrace-Launches-Crypto-Risk-Intelligence-Products-for-Banks.

[45] https://ciphertrace.com/wp-content/uploads/2020/05/CipherTrace_DS-Sentry-20200513a.pdf.

[46] https://www.prweb.com/releases/ciphertrace-introduces-cryptocurrency-real-time-predictive-risk-scoring-to-mitigate-money-laundering-of-crypto-from-theft-and-ransomware-attacks-804594037.html.

[47] https://www.coindesk.com/business/2021/03/11/ciphertrace-releases-its-fatf-friendly-aml-tools-for-crypto-exchanges/.

[48] https://www.binance.com/en/blog/compliance/as-part-of-ongoing-commitment-to-compliance-binance-deploys-ciphertrace-traveler-421499824684902268?lang=en.

[49] https://ciphertrace.com/wp-content/uploads/2022/04/CipherTrace-Inspector-Financial-Investigations-and-Blockchain-Forensics.pdf.

[50] https://fintech.global/2019/01/31/trm-labs-closes-founding-round-on-1-7m/

[51] https://www.newswire.ca/news-releases/trm-labs-the-first-cryptocurrency-risk-management-platform-raises-4-2-million-in-funding-from-initialized-capital-blockchain-capital-paypal-ventures-and-y-combinator-809967429.html.

[52] https://www.trmlabs.com/post/trm-falconx-partnership.

[53] https://www.marketsmedia.com/ftx-us-adds-trm-labs-for-compliance/, https://financialpost.com/pmn/press-releases-pmn/business-wire-news-releases-pmn/xrex-partners-with-trm-labs-to-bolster-platform-security, https://www.trmlabs.com/post/solana-integrates-with-trm-labs-to-offer-enhanced-security-and-compliance-for-

One of the leading European providers of blockchain analytics and crypto compliance solutions Crystal Intelligence introduced real-time risk scoring of transactions in April 2020.[54] The automated blockchain monitoring tool to streamline crypto compliance function, and to detect fraudulent activities and transactions was unveiled more than a year later in August 2021.[55]

Chart 2a reflects the fact that effective real-time and at-scale address and transaction screening and risk scoring tools from blockchain analytics companies were largely unavailable until the middle of 2020. By then, based on Crystal blockchain data, 63% of direct Bitcoin transfers between Hydra and Bitzlato (12,201 BTC out of total 19,456 BTC) had already happened. In a similar vein, Chart 2b demonstrates that automated multi-hop transaction screening tools became available only towards the second quarter of 2021. By then, based on Crystal blockchain analysis, approximately 59% of indirect transfers between Hydra and Bitzlato (3,549 Bitcoin out of 6,005 Bitcoin) had already happened.

In other words, more than half of illicit Bitcoin transfers between Hydra and Bitzlato customers had already taken place before Bitzlato could have access to effective compliance screening capabilities to detect such transfers and to block the appropriate user accounts on its platform.

The Sentencing Memorandum alleges that Mr. Legkodymov was engaged in laundering funds and that Bitzlato "sponsored and enabled it [money laundering]" because "[i]t had the ability to query its customers, to analyze their source and destination addresses, and even to freeze their accounts".[56] However, the Memorandum completely ignores the fact that reliable software tools to screen thousands of transactions and participating wallets were not always available in the private crypto industry or to Bitzlato during the 2018-2022 time period, and that when Bitzlato could detect illicit transfers based on manual review, it did freeze or completely ban hundreds of associated wallets and customer accounts.

V. Bitzlato Compliance Team Worked in 2021-2022 to Build AML Compliance Protocols

Starting from February 2021, when Bitzlato hired Emil Rustemovich as its Chief Compliance Officer (CCO), screening counterparty addresses and cryptocurrency transactions, freezing illicit transfer funds and blocking associated user accounts altogether became one of the top priorities for Bitzlato. This new compliance-centered paradigm could be summarized by a quote from Bitzlato's CCO:

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network-members, https://aithority.com/technology/financial-services/circle-asserts-commitment-to-compliance-with-trm-labs/.
https://financefeeds.com/binance-custody-taps-trm-labs-for-regulatory-compliance-and-risk-management/

[54] https://crystalintelligence.com/news-and-resources/

[55] https://crystalintelligence.com/technology/blockchain-monitoring-with-crystal-a-pillar-for-your-crypto-compliance-and-aml-cft-kyt-processes/.

[56] The Sentencing Memorandum, p.11.

"It is our responsibility not to let users withdraw [funds] to prohibited addresses, to block deposits from prohibited addresses, and to investigate. Over time we will remove from our service those users who regularly send us dirty cryptocurrency, and our own wallets will become clean correspondingly." [57]

But it was a tedious work extending well into 2022, and it took months to start seeing positive results.

1. Efforts to Implement User Verification and Build KYC Framework

Know-Your-Customer (KYC) verification software was developed by Bitzlato and deployed well before the compliance-driven efforts of 2021, but admittedly, it was used by technical support team primarily for account recovery purposes. In 2021, since nearly all of Bitzlato's customers were from Russia, Bitzlato contracted with a Russian company Agnessa to develop software specifically for Bitzlato's compliance needs. Agnessa KYC was designed to replicate verification standards and procedures used by Western customer due diligence providers and specifically modelled after the UK-based company Sum and Substance Ltd. protocols in terms of identification and verifications requirements for new users.[58] [59] It is my understanding from my review of internal Bitzlato communications and conversations with Bitzlato staff that Agnessa KYC software was deployed some time in the second half of 2021.

2. Efforts to Screen Transactions and Deploy Know-Your-Transaction (KYT) Tools

Even prior to deploying real-time address and transaction screening procedures, Bitzlato made substantial effort to freeze funds/block accounts and users associated with Hydra Market. Such efforts were implemented in various ways. Users could be blocked/funds frozen as the result of investigations by technical support team. Bitzlato blocked users with Telegram nicknames/account names containing words "Hydra", Scam", or names of known drug stores in Moscow. Bitzlato also checked transaction ticket descriptions for words containing "Hydra" or drug-related uses of crypto funds.

Shortly after the new CCO Emil Rustemovich joined Bitzlato, in April 2021 he onboarded VALEGA Chain Analytics.[60] Valega is a Finland-based blockchain analytics company providing wallet and transaction screening services to assist virtual asset service providers in combatting risks of illicit finance and money laundering. From then on, Bitzlato obtained a capability to flag risky addresses and/or transactions on either real-time or retrospective basis based on Valega's automatic risk score determinations. Additionally, Bitzlato was now able to proactively verify large deposits or withdrawals of cryptofunds

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[57] Emil on Slack conversation thread (2021-12-23 19:57:47). Native language quote: "Наша ответственность - не позволять пользователям совершать выводы на запрещенные ресурсы, блокировать депозиты с запрещенных ресурсов и проводить разбирательство. Со временем мы очистим сервис от пользователей, которые регулярно нам льют грязную криптовалюту и соответсвенно очистятся наши кошельки."

[58] See https://sumsub.com/.

[59] In the second half of 2021, Bitzlato seriously considered and tested Sumsub KYC application both for mobile users and users of the Bitzlato exchange website in November-December 2021, but ultimately decided against this service due to its prohibitively high cost.

[60] https://www.valegachain.com/. See "Welcome to Valega" letter dated April 19, 2021.

against Valega's database of "blacklisted" crypto wallets associated with dark web, scams, hacks, theft, etc.

Bitzlato's compliance team efforts to implement robust KYT protocols intensified in the Fall of 2021. Starting from early November 2021, the team also started building an internal database, dubbed "Meduza", to store the risk-scoring data on addresses and transactions obtained from Valega.[61] This way, Bitzlato could complement real-time Valega determinations with additional screening against the ever-expanding internal repository of "dirty addresses" and suspicious transactions.

In November 2021, Bitzlato, using Valega service, screened both source addresses and transactions for deposits and destination addresses for withdrawals. Transactions were screened for USDC, USDT, BTC, LTC, DASH, BCH, albeit the vast majority of volume took place in BTC. Bitzlato faced a very challenging task, as in addition to screening real-time crypto transactions, it also had to verify users, prioritize blocking those users with failed verifications, as well as to screen a huge number of past transactions.

For example, in an effort to ringfence itself from suspicious addresses, in February 2022 the Bitzlato compliance team started screening transactions for the preceding six months, and by the end of that month 314,000 transactions were still in the backlog to be screened. As of the end of February 2022, 20% of screened transactions during the preceding three months were red-flagged transactions, and out of those 76,159 transactions were labelled as "Hydra Market." As of April 8, 2022, 500,000 of transactions had been screened by the Bitzlato compliance team out of the total of 700,000 past transactions subject to screening.

3. Implementation of Full Customer Verification and Robust Compliance Protocols in April 2022

The preceding multi-month tedious work on deploying KYC protocols, at-scale address and transaction screening capabilities based on Valega services and internal database resources, as well as compliance screening of hundreds of thousands of prior transactions, ultimately allowed Bitzlato to transition to a full customer verification system coupled with a robust AML compliance screening protocol. In a nutshell, by April 7, 2022, Bitzlato's implemented the following compliance policies and procedures: (1) Server/service verification to implement full user verification on Bitzlato platform; (2) Automatic account blocking for incoming transactions from dark markets based on Valega information; (3) Automatic blocking of withdrawals to darknet markets based on Valega determinations.[62]

More detailed steps of Bitzlato AML compliance protocol as implemented in early April 2022 and approved by Anatoly Legkodymov involved the following. If Valega determines that a user's transaction/wallet is

[61] See https://github.com/bitzlato/meduza for open-source code.

[62] Starting from April 8, 2022, withdrawals to dirty addresses were screened and blocked automatically for addresses risk level 3 with risk confidence 3 which was later expanded to risk level 3 with risk confidence 1. In late May 2022, Bitzlato put together specifications for an initiative that would add AML checks for users of the platform. Valega was the intended service provider to perform such checks. See Slack Thread Reply, Alona Zahorulko 2022-05-27 17:37:59, "готово описание проекта по добавлению aml-проверки для юзеров", https://www.notion.so/woobit/aml-ccca67d9b9b44c77ba737caa6968580c.

linked to a prohibited resource, then: (1) the user is blocked with an indication of the reason (2) an account ownership is being checked (whether the account was compromised to commit fraud); (3) proper identity verification check (identity verification + proof of residence); (4) If the user is blocked because of the deposit, then a source of funds is requested, the user must show from which resource the operation was initiated. If the block is for withdrawal, then the user must explain the purpose of sending funds and provide the resource where the wallet belongs to; (5) Based on the data provided, a decision is made: warning and unblocking or termination of using the platform, refund and ban; (6) On the basis of the investigation, a report on a suspicious transaction is created, which includes the information obtained during the investigation.[63]

4. Bitzlato AML Compliance Development Efforts Were Significantly Hampered by Persistent Technical Challenges and Other Factors Out of its Control

Bitzlato encountered substantial and recurring technical difficulties, and consequently material delays, with rolling out robust KYC and transaction/address screening capabilities through Spring of 2022 for a variety of reasons. For example, efforts to block users based on AML screening at the end of February 2022 were disrupted due to the fact that 70% of Bitzlato's customer service staff was in Ukraine which had been just invaded by Russia.[64] Meduza internal transaction screening database was overwhelmed by the combination of old transactions submitted for verification and new incoming deposit and outgoing withdrawal transactions. Bitzlato team also had to systematically engage in a manual review of transactions: as of April 8, 2022, 3,309 transactions/wallets had been manually screened, of which 1,950 were found to be "dirty".

More substantially, Valega's screening and timely response capabilities turned out to be a true bottleneck for Bitzlato until the middle of April 2022. Valega was founded only in 2019, with its first seed investment in September 2020. For the first 12 months after onboarding with Bitzlato, Valega address and transaction screening services were subject to multiple system revamps, security-related interruptions of up to 10 hours and a restrictive limit of just one API screening call in every 10 seconds.

[63] See Emil, Slack Reply Thread, 2022-04-06 16:55:12. "Если Valega определяет, что транзакция/кошелек пользователя связан с запрещенным ресурсом, то:

1. Пользователь блокируется с указанием причины
2. Проводится проверка на предмет владения аккаунта (не был ли аккаунт взломан для совершения мошенничества)
3. Проводится надлежащая проверка личности (верификация личности + подтверждение места жительтсва)
4. Если блок за депозит, то запрашивается источник средств: пользователь должен показать с какого ресурса совершил операцию. Если блок за вывод, то пользователь должен объяснить цель отправки средств и предоставить ресурс, которому принадлежит кошелек.
5. На основании предоставленных данных принимается решение: предупреждение и разблокировка либо прекращение сотрудничества, возврат средств и бан.
6. На основании разбирательства формируется отчет о подозрительной транзакции, который включает все данные разбирательства."

[64] See Slack Thread, 2/25/22 20:03:36.

Based on the number of transactions per day, Bitzlato needed a robust and fast ability to batch send transactions and addresses to Valega for compliance screening. Bitzlato needed an AML screening speed of at least 2-3 addresses/transactions per second, while it could take up to 5 minutes to get one address response from Valega. Furthermore, at least initially, Valega limited batch requests from Bitzlato to no more than 10 addresses per one second. Valega would also often provide only a partial response to batch requests containing 10 addresses and transactions. Bitzlato team diligently worked with Valega to remove limits on the number of API calls to screen transactions and response delays, but, based on my review of internal communications, many transactions submitted by Bitzlato were not screened by Valega due to API call timeouts. Bitzlato developer team also complained that some address/transaction requests to Valega would come back with an indeterminate response "false."[65]

Equally, if not more importantly, Bitzlato's efforts to deploy a speedy and robust AML transaction monitoring system were hampered by the refusal of established blockchain analytics and AML solution providers Chainalysis and Ciphertrace to provide transaction and address screening services to Bitzlato. Bitzlato internal communications confirm such refusals.[66]

My review of external Bitzlato communications also indicate Chainalysis' and CipherTrace's refusal to onboard Bitzlato as the latter was looking to rectify its compliance protocols. In particular, newly hired Chief Compliance Officer Emil Rustemovich contacted the US blockchain company CipherTrace at the end of March 2021, i.e. very shortly after he joined the platform in February 2021. Having scheduled the initial conversation, CipherTrace abruptly cancelled it. In response to Bitzlato CCO's attempt to reschedule, the response from CipherTrace was clear and blunt: "Unfortunately not Emil ... too much risk"[67]. As the result of this refusal, Bitzlato was forced to subscribe to the blockchain screening services of much less established and less reliable Finland-based provider Valega Chain Analytics OY.

Later on, Bitzlato CCO reached out to Chainalysis on November 9, 2021, with an explicit objective to onboard the second crypto compliance service provider to "double KYT additional verification with your [Chainalysis] service" and "to cooperate with you [Chainalysis], to reduce compliance risks." [68] The CCO also explicitly stated that Bitzlato wanted to double check each of the 150,000 transactions for deposits and withdrawals per day that the platform had at that time[69]. After several reschedulings and delays of

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[65] See, for example, Slack Reply Thread, @Emil valega весь день отдает
I, [2022-04-15T19:27:50.092345 #50676] INFO -- : Make request realtime_risk_monitor/risk/analysis address_transactions=["bc1qkuctudkckdyffutkzg087uzmgqptkrlhvppjhw", "19NrqVCVD5tsWqSEHHfnw3BdRRfe8r9o9n", "1HuaiwaDnvDfpTHv7kW7qypdhhv7J5QHFj", "3AVGgCzjtFVhC2LC8P8HVaEZZVUUADkvYF", "bc1qcwzd9jpsa3a6lqqzl5ky90nyk6nueu5d0pvww0", "3BqNBz9GZ8jmTbpYHbzz8sC5ebcM34SSjB", "3Lqg9igE99d3Y58TptdfauZrRKZm6drMSS", "32eUDv8rrYWHBdCb8AvNxjJKYSE7rrUzFo", "bc1qczxked8mncseyfxdgyny307z9ytsveeeyz2rl7", "1KxrMM85asNjaSW7CCabMoqohthD7JjNKJ"]
I, [2022-04-15T19:27:50.433643 #50676] INFO -- : ValegaClient status=200 body={"result":false}

[66] See Slack Thread by Emil Rustemovich as of 4/17/22 13:03:52, *Thread Reply:* ЧА не хотят сотрудничать, ciphertrace отказались сотрудничать, coinfirm мы отказались, кристал хотят 500К в года за безлимит, выше подсказали scorechain, посмотрю что за сервис." Translation: "CA [Chainalysis] does not want to work with us, Ciphertace refused to collaborate, we decided against confirm, Crystal wants 500K for unlimited access, earlier I was suggested Scorechain, I am going to check this service out".

[67] Communucation between Emil Rustemovich and Paul Reyff as of March 29, 2021.

[68] Email from Emil Rustemovich to Karim Halabi of Chainalysis as of November 9, 2021, 15:36.

[69] Email from Emil Rustemovich to May Ngiam of Chainalysis as of November 18, 2021, 11:31.

the product demo, Chainalysis staff went radio silent at the end of November 2021 despite the repeated attempts of the Bitzlato CCO to conduct the promised product demonstration.

5. Despite Technical Challenges, AML Compliance Protocols Did Make a Big Positive Difference for Bitzlato Before It Was Shuttered in Early 2023

Notwithstanding all technical and service provider-related difficulties, Bitzlato team actively worked in 2021-2022 to reduce the volume of illicit activities on its platform and to build a robust wallet and transaction screening system for compliance purposes. By October 2022, Bitzlato had 6 people on its staff dedicated to compliance.

From 2018 to 2023, the Bitzlato technical support team completely banned (disabled accounts of) 31,768 user accounts from its services for fraudulent activities, suspicious activities, fake documents, fake verifications, hacks and scams. Out of the total, more than half of bans occurred in 2021 and 2022, with 2021 alone accounting for over 15,000 of user account bans.[70]

Compliance-related account freezes by technical support also ramped up sharply in 2021 and 2022. Out of total 11,264 accounts frozen by Bitzlato technical support team between January 2020 and January 2023, 2,529 account freezes in 2021 and 2022 were specifically due to suspected AML violations and compliance checks. Moreover, out of AML-related account freezes by the technical support, 297 compliance account freezes in 2021 were carried out because Bitzlato detected the word “Hydra” either in users’ Telegram names, or deposit and withdrawal tickets. [71]

As Bitzlato worked on implementing real-time screening of incoming and outgoing cryptocurrency transactions, as well as deposit and withdrawal addresses, using Valega screening services, the AML-related account freezes by the Bitzlato’s compliance team (i.e. freezes in addition to the technical support team freezes) ramped up as well. Chart 5 shows that such AML screening-based account freezes skyrocketed from practically zero in the first half of 2021 to 379 in the second half of 2021 to 14,078 freezes in 2022. [72]Compliance-related complete user account closures by the Bitzlato compliance team also sharply increased from 92 in 2021 to 1,232 in 2022.[73]

My examination of Bitzlato records showed that, in addition to automatic address and transaction screening, between January 2021 and January 2023 (all prior to the involvement of law enforcement) the internal compliance team conducted at least 1,039 investigations (Suspicious Activity Reports).[74] As an example of such an investigation, the account of user #12001439 was frozen on April 30, 2022 and eventually fully blocked, with the reason recorded as “Withdrawal to a forbidden resource”. My further analysis showed that the user has been withdrawing Bitcoin to address 3G2M2ChUCuA13i7gyZSC155xNH8hhrzpRs. This address was flagged by Bitzlato as “forbidden resource” because on February 5, 2022, it received 0.01097 BTC from Hydra addresses and then sent them to Bitzlato

[70] See Chart 3.
[71] See Chart 4.
[72] Chart 5.
[73] Chart 6.
[74] See file “SAR Registry.xlsx”.

address 15SsAkX88icpKQTd7qDdkRx2d5jwVNmhq5[75]. In other words, Bitzlato compliance team blocked this user precisely because it interacted with Hydra Market. Based in internal Bitzlato's SAR registry, Chart 7 demonstrates that the number of SARs resulting in a full block of user account rose more than 26-fold from 20 in 2021 to 531 in 2022.[76]

Efforts of the Bitzlato team to build and implement robust AML/KYC screening protocols in 2021 and 2022 resulted in a significant reduction in the amount of illicit transactions on its platform. Chainalysis blog post on January 18, 2023, included a chart showing that the share of illicit flows at Bitzlato platform, after peaking in 2021 at more than 1/3 of overall activity, sharply fell in 2022 to around 10%[77]. Additionally, based on the data of the internal compliance team, "dirty" BTC deposits into Bitzlato wallets declined from 11,342 in July 2021 to only 173 deposits in August of 2022. Chart 8 demonstrates that, proportionally, "dirty" BTC deposits into Bitzlato wallet declined from 19.5% of overall deposits in July 2021 to 0.5% in August 2022.[78] Such decline occurred independent of the shutdown of Hydra Market in April 2022 because "dirty" BTC deposits into Bitzlato wallets had declined dramatically and consistently already in the first quarter of 2022.

Bitzlato also actively responded to the inquires of law enforcement authorities of various countries. Bitzlato responded to approximately 1226 requests from the Russian and Belarussian authorities. It also provided information with respect to approximately 334 requests from the foreign authorities, including requests from the following countries: Austria, France, Germany, Ukraine, Italy, South Korea, Ireland, Belgium. Hungary, Romania, Netherlands, Switzerland, Spain, Czech Republic, Poland, Australia, Bulgaria, Estonia, Lithuania, Portugal, the UK, and the United States. Between June 2021 and early January 2023, Bitzlato responded to at least 36 inquiries from various branches of the US Government.

Based on my review of the case materials, Bitzlato cooperated with and responded to inquiries from various law enforcement agencies in the United States starting from June 2021. These responses included 16 inquiries from the Federal Bureau of Investigations, 6 inquiries from the Department of Homeland Security, 13 inquiries from the offices of the United States Attorneys of Maryland, Virginia, Illinois, New Jersey and Washington, and one request from a local police department in Massachusetts. I have reviewed Bitzlato's detailed responses to the US law enforcement inquiries which included spreadsheets with detailed information about a user's ID, Telegram name, email address and history of its transactions on Bitzlato P2P platform, as well as deposits and withdrawals of cryptocurrency. Ironically, the last response by Bitzlato to an FBI inquiry occurred on January 6, 2023 – just eleven days before Mr. Legkodymov' arrest in the United States.

[75] BTC transaction hashes 5b21587ed8216f62cd7a853e1a34283ef7277f47083c8e094f8162a44c5ad4a5 and 37ac362b647aaa1e8d234a188c37dda4edffb85442cd2c6dc0845fdea3ee7f2e.
[76] Chart 7.
[77] https://www.chainalysis.com/blog/us-authorities-shut-down-high-risk-exchange-bitzlato/.
[78] See Chart 8.

VI. Mr. Legkodymov Did Not Deliberately Attempt to Avoid Implementing AML Policies – In Contrast, He Pushed His Staff Repeatedly for a Faster Implementation of Such Policies.

My review of Bitzlato records and communications shows that, contrary to a person who would seek to conspire or cause his company to conspire to engage in money laundering, Mr. Legkodymov persistently pushed his staff to implement compliance controls and clean Bitzlato wallets once he learned about the extent of the platform's exposure to illicit cryptocurrency flows.

1. Bitzlato Was Not Subject to Anti-Money Laundering and Counter Terrorist Financing Ordinance in Hong Kong

As a peer-to-peer platform, facilitating exchange of virtual assets for fiat money, Bitzlato could be deemed to be money service operator in Hong Kong. However, since neither Bitcoin is a security, nor were any other assets traded on the platform recognized as securities by the Hong Kong regulations, Bitzlato did not fall into the category of Virtual Asset Trading Platforms and thus was not subject to the Anti-Money Laundering and Counter Terrorist Financing Ordinance (AMLO) enforced by the Securities and Futures Commission (SFC) as part of licensing regime for VATPs.

Bitzlato, with its almost exclusive user Bitcoin-for-fiat focus, could only be classified as Virtual Asset Service Provider (VASP). On 3 November 2020, the Financial Services and the Treasury Bureau (FSTB) published a consultation paper on its proposals to amend the AMLO to bring exchanges which offer crypto assets that do not qualify as "securities" within the regulatory oversight of the SFC (VASP regime). The new rules were designed to close the existing regulatory gap and regulate VASPs like Bitzlato for the first time. The FSTB published its consultation conclusions on the VASP regime in May 2021 and an amendment bill was expected to be introduced into the Legislative Council in the second quarter of 2022. In reality, the Hong Kong Legislative Council passed the Anti-Money Laundering and Counter-Terrorist Financing (Amendment) Bill 2022, establishing a licensing regime for VASPs, only on December 7, 2022. The new law required VASPs to apply for a license with the SFC to operate in Hong Kong and imposed statutory AML and CTF obligations. Similar to traditional financial institutions, licensing would have required Bitzlato fitness and properness, financial resources, and risk management policies and procedures to be established in place. On May 25, 2023, the SFC published AML/CTF Guidelines for SFC-licensed VASPs, which mandated that VASPs comply with the new licensing regime by June 1st, 2023. Bitzlato, however, which had worked in 2021 and 2022 to install proper AML policies and procedures and which would have been able to be in compliance with the Hong Kong new AMLO regime for VASPs, had been shut down by the international law enforcement authorities before the new Hong Kong requirements went into effect.[79] In summary, registered as a Hong Kong company, Bitzlato was not subject to and did not violate AMLO regulations of this country for virtual asset service providers and was working toward improving its compliance and updating with changing technology to further its goal of matching industry standards.

[79] In Europe, the 5th Anti-Money Laundering Directive (5AMLD), which was created to prevent money laundering and terrorist financing, and whose scope was expanded to include virtual currency exchanges for the first time, went into effect in January 2020.

Nor was Bitzlato accused of violating US FinCEN registration requirements for money service businesses (MSB) along with the associated obligations of registered MSBs to develop and implement KYC and AML/CFT policies and procedures under the Bank Secrecy Act. As I discuss below, Bitzlato never had any material exposure to US residents, serving overwhelmingly users from Russia and former Soviet Union jurisdictions.[80]

Based on codes of conduct, published by the Association for Digital Asset Markets ("ADAM"), Global Digital Finance ("GDF") and the Global Digital Asset & Cryptocurrency Association (the "Global DCA") in 2019 and 2021, there are several key common elements required of robust AML/KYC policies and procedures: (1) written AML and KYC policies/procedures reasonably designed to prevent the business from being used for money-laundering and/or terrorism financing, (2) appointment of a compliance officer; (3) ongoing AML and KYC training for employees; (4) regular risk assessments of both transactions and users; (5) investigations and reporting of suspicious activities and/or transactions to the appropriate law enforcement authorities; (6) maintenance of appropriate records. In 2021-2022 Bitzlato was working actively to effectuate compliance in categories (2), (3), (4) and (5) (in cooperation with Valega), as well as category (6) (Meduza database). Regarding compliance with category (1) requirements, Bitzlato had AML manuals and procedures since at least 2020. This is despite the fact that for most of time, Bitzlato operated as P2P trading platform, not as a centralized crypto exchange. The fact that Bitzlato lacked KYC policies between 2016 and 2020 does not seem unusual, particularly given that Bitzlato's exchange business was crypto-to-crypto and P2P was fiat-to-crypto Perhaps equally importantly, proven at-scale commercial tools to assess the risk of cryptocurrency transactions in real time were largely unavailable for most of this time period.

2. Mr. Legkodymov Pushed His Staff Repeatedly to Implement AML Compliance Policies and Procedures

Legkodymov was pushing his compliance staff for a universal user verification, as well as verification of all suspected "dirty" transfers since at least late February, 2022.[81] His goal was to ensure that "there is no more dirt, up to zero and everybody is verified."[82] He also looked to establish policies by which a Bitzlato user could only deposit his own funds, and Bitzlato verified the source of funds in accordance with European bank practices."[83]

Given the painfully slow progress of such efforts, Legkodymov was increasingly frustrated with his team to the point that in late February 2022 he was "ready to shut down the company", "place a veil on the [exchange] website and in the bot" and return money [to users] upon requests."[84] He repeatedly complained that "we are not going according to the plan promised to me to get rid of dirty [activities]"[85]. He also tried to understand why the old user verification service could not cope with the challenges of

[80] "A foreign-located business is an MSB if it does business "wholly or in substantial part within the United States. (31 U.S.C. §§ 5312(a)(6), 5312(b), 5330(d); 31 C.F.R. § 1010.100(ff)).
[81] See Slack Conversation at 2022-02-25 20:04:45 onwards.
[82] *Id*., 2022-02-26 @ 15-05-22.
[83] 2022-02-26 @ 22:32:48.
[84] *Id*., 2022-02-25 @ 20-25-47.
[85] *Id*., 2022-02-25 @ 23:34:15

intended global user verification and demanded explanations from his colleagues. Later on, as the efforts of Bitzlato compliance team progressed, in early April 2022 he approved the 6-step AML screening procedure proposed by Bitzlato CCO in conjunction with additional screening and blocks by the technical support team of users related to keywords referencing Hydra, controlled substances, etc. [86]

On April 7, 2022, Legkodymov created a draft proposal to crypto exchanges for an automatic exchange of transaction hash data and hashed customer KYC data with such exchanges to enable them to screen and risk-score crypto transfers faster for AML purposes.[87] In suggesting such a data-sharing proposal, he was looking to leverage user data collected via the Bitzlato KYC portal Agnessa discussed earlier.

On the same day Legkodymov issued a stern demand to his compliance team on behalf of Bitzlato's key stakeholders to combat darknet transactions: "If a user attempts even a single withdrawal to the dark market, we fully block his account, return the funds to his "clean" wallet, and deny him any further service."[88]

Additionally, in early April 2022, he chastised his compliance colleagues, including the Bitzlato CCO, for not addressing his requests which he had made one month before and had been asking for almost every week, for instance, a report on withdrawals to darknet wallets for the preceding three months, based on Valega onchain transaction analytics. Quoting Legkodymov from Bitzlato internal conversation records, "I am already beginning to perceive this as a criminal inactivity or sabotage. Please make the work regarding AML as transparent as possible, otherwise I will be forced to simply suspend the service. I am dead serious."[89] Legkodymov also routinely asked for but did not receive from his staff audit reports on security-related issues: transaction audit for the preceding six months, number of government agency requests, number of requests from special services, number of verified user accounts, and information on transactions with relationship users.

Mr. Legkodymov was also regularly monitoring monthly reports on the number of "dirty" transactions and verified with the team that the corresponding deposits has been blocked.[90]

As Bitzlato was strengthening its compliance and screening procedures over the course of 2022, it started sharing its accumulated knowledge and data with other blockchain analytics and compliance services providers. For instance, as Bitzlato switched from Valega to Scorechain to improve scalability and reliability of screening capabilities, it shared its information on darknet wallets with Scorechain, some of which were not known to Scorechain and would have gone undetected if not for Bitzlato's effort.[91] By then Bitzlato's internal database Meduza, storing address and transaction screening results, had 19,905 addresses labelled as "DARK MARKET", including 17,559 wallets specifically identified as "Hydra Market". For example, in an email to Scorechain on December 21, 2022, Bitzlato staff alerted the former about

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[86] See Emil, Slack Reply Thread, 2022-04-06 16:55:12.
[87] *Id.*, 2022-04-07 @ 16:27:13.
[88] *Id.*, 2022-04-07 @ 19:15:54.
[89] *Id.*, 2022-04-07 @ 19:42:08.
[90] Id., 2022-04-10 @16:29:43, 2022-06-02 13:45:01,
Thread Reply: Всех заблочили? (Were all these blocked?)
Andrey P 2022-06-02 13:45:32
Thread Reply: да, конечно (Yes, of course).
[91] https://www.scorechain.com/products/crypto-wallet-and-transaction-screening.

certain addresses from three darknet markets which were not detected by the screening tools of Scorechain.[92]

It is ironic that the Sentencing Memorandum extensively cites to Mr. Legkodymov's internal communications with his staff and his express disappointment regarding Chainalysis' February 2022 analysis of Bitzlato's exposure to Hydra as the "proof" of Mr. Legkodymov's knowledge of such exposure and his willful avoidance of establishing anti-money laundering controls.[93] The Sentencing Memorandum, however, fails to put these communications into a full context and to mention that the same Chainalysis had stonewalled Bitzlato's attempt to use its wallet and transaction screening services in November of 2021, forcing the latter to resort to services of less established and less reliable provider Valega to clean the illicit finance exposure. Nor does the Sentencing Memorandum mention the fact that a few months earlier in March 2021, Bitzlato was refused screening services by another established U.S. blockchain analytics company CipherTrace. If anything, Mr. Legkodymov's frustration with the February 2022 Chainalysis report was indicative of Bitzlato's struggles to effectively ramp up their compliance protocols, not of his deliberate avoidance of such protocols.

VII. Bitzlato Did Not Solicit US Customers and Had a Miniscule Exposure to the United States, If Not Zero Exposure

Bitzlato's Terms of Service explicitly prohibited residents of the United States from using the services on its platform, and a prospective user had to click and accept the Terms of Service to register an account on Bitzlato.[94] At the same time, due to the fact that users often accessed platform via Telegram, Bitzlato's ability to block users based on their IP addresses was limited. To address this technical issue, Bitzlato routinely collected information on a user's fiat payment system which was available due to the fact that Bitzlato peer-to-peer platform only allowed exchanges of cryptocurrencies for fiat money and vice versa. Bitzlato's consistent effort to exclude US residents from its platform is fully corroborated even if one were to accept at face value the alleged $2.4 million transacted by customers located in the Eastern District of New York with Bitzlato.[95] For such alleged transaction volume represents a mere 0.05% of the overall transactional volume of $4.58 billion claimed to be processed by Bitzlato between 2018 and 2022. In other words, Bitzlato purposedly avoided the US customers and had zero footprint in the United States, save for selected US users who deliberately violated the Terms of Service at their own will.

Bitzlato monitored activities of its users including those whose residence could not be verified but who used US payment system for at least a fraction of their fiat transactions. I have reviewed such tracking data which includes 229 unverified users and came to the same conclusion that Bitzlato potential exposure

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[92] Email dated 12/21/2022, "We are in the process of switching from our current AML service to yours. The thing is that many darknet wallets such as OMG (1C942pYhz1iEVtscedac6vPkcAUfdMA2WX), Blacksprut (35zuWiEJs87xDqcE6sATsGtWw4rhsbEXc), Matanga (14CZpSRx6LCTrDGNCcYchmxRYnGd6rzEcG), your system is not detected. How can we solve this problem?".

[93] Sentencing Memorandum, pp. 5-6.

[94] Bitzlato Terms of Service.

[95] Amended Affidavit and Complaint, Para. 36.

to US customers was miniscule if not zero.[96] Figure 2 breaks such users into groups by fiat currency used to either purchase cryptocurrency or to sell it for, or by the geographical location of their IP addresses whenever available.

First, as Figure 2 demonstrates, there are no transactions where USD was used. Second, Bitzlato services were used overwhelmingly by Russians and residents of other republics of the former Soviet Union. Specifically, 69.1% of transaction volume and 76% of user count are attributable to Russia and other parts of former Soviet Union based on the combination of fiat currencies used (RUB, Ukrainian Hryvnia UAH, Kazakhstani Tenge KZT, etc.) and their Russian IP addresses. In fact, given the popularity of AED (United Arab Emirates Dirham) among Russians to purchase cryptocurrencies, it is highly likely that Russians and former Soviet Union jurisdictions account for 99.1% of Bitzlato transactions by unverified users and 82.5% of Bitzlato unverified users.

Third, users with unverified residence, while accounting for 15.7% of the unverified user count, consummated only 0.8% of the unverified-residence transaction volume on Bitzlato. Finally, the cohort of hypothetically US-based unverified users, i.e. those with unverified residence and unknown IP address, account for mere 0.4% of the unverified transaction volume. However, even this 0.4% of transaction volume may well be unrelated to the United States because there is no evidence to confirm that such users are in fact US residents.

I have also reviewed several files with user trading activity, provided by Bitzlato, which contain records for a total of 49,937 trades during one day in July of 2019, 2020, 2021 and 2022.[97] When all trades are considered (both cancelled and confirmed), only 0.16% of trades by count and 0.17% by market value involved exchanging cryptocurrencies for USD (79 trades with the total value of $20,805). When 36,102 confirmed customer trades are separately considered, only 0.14% of trades by count and 0.13% of trades by market value involved exchanging cryptocurrencies for USD (only 41 transactions with the total value of only $9,222).

A related implication of the above findings is that since Bitzlato was a foreign company with miniscule to no exposure to the United States, it was never a money transmitter business (MSB) subject to FinCEN registration requirements and associated AML/CFT obligations under the Bank Secrecy Act.[98]

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[96] See file "Unverified users (American PS).xlsx

[97] See files 2019-07-22.xlsx, 2020-07-22.xlsx, 2021-07-22.xlsx, 2022-07-22.xlsx.

[98] "A foreign-located business is an MSB if it does business "wholly or in substantial part within the United States. (31 U.S.C. §§ 5312(a)(6), 5312(b), 5330(d); 31 C.F.R. § 1010.100(ff)).

VIII. The Extent of Bitzlato's "Wrongdoings" and its Exposure to Illicit Finance Was Much Less Than of Other Cryptocurrency Exchanges Whose CEOs Were Subject to Criminal Prosecution in the U.S.

This section of my report highlights the fact that Bitzlato's exposure to illicit finance was much less than that of some much larger crypto currency exchanges whose executives and control persons were criminally prosecuted in the United States. I also show that Mr. Legkodymov and Bitzlato's wrongdoings are far smaller in scope compared to those of their larger and more consequential counterparts. In particular, I compare Bitzlato and Legkodymov to Binance and its former CEO Changpeng Zhao, as well as BitMEX and its former CEO Arthur Hayes. The crypto billionaire Mr. Zhao was sentenced to only four months in prison on April 30, 2024[99], while Mr. Hayes's sentence in May of 2022 was limited to six months of house arrest and two years of probation.[100] For context, Binance is the largest crypto exchange in the world, with over $124bn in user assets and $15.9bn in spot daily trading volume and $42.2 billion in crypto derivatives daily trading volume, whereas BitMEX user assets and daily crypto derivatives trading volume are $1bn and $375 million, respectively.[101] Bitzlato, with its cumulative transaction volume of $4.58bn during the entire May 2018 – April 2022 period, effectively processed over its entire history less than 1/3 of Binance's daily spot trading volume, and is truly a drop in the bucket in comparison to either of these two centralized exchanges.

To avoid any suspicions of bias in my comparisons, they are solely based on allegations, claims, data and information contained in the DOJ, CFTC and FinCEN pleadings and other government documents directly related to criminal and civil charges levied against Binance, BitMEX and their control persons. Wherever applicable, government claims or facts are referenced to the appropriate source document(s). See Figure 3 for details.

1. Bitzlato's Exposure to Illicit Finance Was Far Less Than That of Binance and BitMEX, and, Unlike These Two Large Crypto Exchanges and Their CEOs, Mr. Legkodymov and Bitzlato Did Not Seek to Avoid AML/KYC Requirements

Regarding the volume of transfers of Bitcoin with Hydra Market, Figure 3 shows that, despite all the focus of legal authorities and the public media, Bitzlato pales in comparison with Binance. Specifically, based on the analysis by Crystal Intelligence, between May 2018 and April 2022 Bitzlato wallets "processed" less than 1/3 of Binance's transactional volume with Hydra wallets when measured in terms of BTC count (estimated 25.5K BTC versus 80.0K BTC, or 31.8%), and less than half of it when measured in terms of transferred market value (estimated $449 million versus $991 million, or 45.3%).

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[99]https://www.cnn.com/2024/04/30/business/binance-founder-sentenced-money-laundering/index.html#:~:text=Changpeng%20Zhao%2C%20the%20founder%20of,effective%20anti%2Dmoney%20laundering%20program.

[100] https://www.wsj.com/articles/former-bitmex-ceo-sentenced-to-house-arrest-on-anti-money-laundering-charges-11653086530

[101] https://www.coingecko.com/en/exchanges/bitmex#futures, https://www.coingecko.com/en/exchanges/derivatives, https://www.coingecko.com/en/exchanges/binance, https://defillama.com/cex/binance-cex#information, https://defillama.com/cex/bitmex#information.

Second, Bitzlato wallets did not transact with U.S. or U.N. sanctioned entities or wallets engaged in terrorist financing, while the addresses of the other two exchanges did. Based on the US Government claims, regulatory pleadings and other documents, Binance wallets engaged in nearly $900 million of transactions with Iranian entities, transacted with sanctioned jurisdictions of Cuba, Syria, Ukrainian areas annexed by Russia, and Russia itself after the start of the war against Ukraine, as well as several terrorist group-related wallets. BitMEX addresses interacted with wallets from sanctioned jurisdictions of Cuba, Iran, Syria, North Korea and Sudan.

Third, Bitzlato and Mr. Legkodymov did not target or solicit US-based users and did not derive any profits from their transactions. Its customer base was overwhelmingly from Russia or countries from the former Soviet Union. If, as claimed, but not substantiated, by the US Government, some US users did interact with Bitzlato wallets, they did so in circumvention of the explicit prohibition in the Bitzlato Terms of Service for US users to access its platform in an environment where, unlike centralized web-based crypto exchanges Binance and BitMEX, Bitzlato had limited capacity to track its users' IP addresses and their geographical location. Bitzlato was not a US money transmitter subject to the Bank Secrecy Act and its implementing regulations, and it was never accused by a U.S. Government agency of violating such regulations.

In stark contrast to Bitzlato, in accordance with the U.S. Government and regulatory agency claims, both Binance and BitMEX, including their control persons, actively targeted and solicited US residents for business and, in fact, made US-based "VIPs" a pillar of their commercial profitability. As Figure 3 demonstrates, such efforts are extensively documented and claimed in the US Government and regulatory pleadings, consent orders and agreements. Binance derived 20-30% of its business from the United States, processed billions of dollars in transactions with US users, and had millions of US accounts. The exchange made more than $1.6bn in profit from the US customer base. BitMEX, on the other hand, had 4-8% of its customer base from the Unted States, had more than $10bn in BTC deposits from the US users, amassed tens of thousands of US user accounts, and made $1bn in profit from US-related transactions.

Lastly, while Mr. Legkodymov knew about the exposure of Bitzlato platform to Hydra Market, he actively pushed his compliance team in 2021-2022 to implement a robust KYC/AML program and to clean Bitzlato wallets of "dirt." As a result, Bitzlato achieved marked improvements in the quality of its compliance screening protocols and was on the path to further steady and consistent improvements in compliance protocols until it was shuttered in early 2023. There is no evidence that I have reviewed that Mr. Legkodymov ever willfully or deliberately avoided implementing KYC/AML programs of any jurisdiction.

Again, in stark contrast to Mr. Legkodymov and Bitzlato, as per the US Government and regulatory claims, both Binance, BitMEX and their respective former CEOs Changpeng Zhao and Arthur Hayes not only knew about the substantial exposure of their exchanges to various forms of illicit finance and the accompanying KYC/AML violations, but they and their management teams also made conscious effort for multiple years to avoid implementation of KYC/AML controls to maintain their profitability. Not surprisingly, both were charged with "willfull violation" of the Bank Secrecy Act and willful failure to implement AML programs. There is truly no comparison between willlful violation of AML regulations and Mr. Legkodymov's stance, even if one were to accept as true what the Sentencing Memorandum described as "passive disapproval" on his part of slower than expected progress of Bitzlato compliance team to install effective wallet and transaction controls.[102]

Binance's deficiencies in dealing with illicit cryptocurrency flows were known as early as the beginning of 2020. In a January 2020 analysis blog post, Chainalysis traced more than half of $2.8 billion in Bitcoin that moved from criminal entities in 2019 to just two exchanges - Binance and Huobi - more than any other cryptocurrency exchange.[103] Binance received $770 million in illicit Bitcoin funds in 2019, according to Chainalysis (27.5% of $2.8 billion)[104]. The blockchain analytics firm also concluded that the majority of illicit inflows to Binance came from "rogue" over-the-counter (OTC) crypto brokers some of whom specialized in providing money laundering services to criminals. Binance's problematic approach to compliance and anti-money laundering controls has also been widely publicized and evident through multiple adverse regulatory actions in various countries.[105]

The US Department of Justice described the past behavior of Binance - the largest crypto exchange in the world - as "turning a blind eye to its legal obligations in the pursuit of profit." In November 2023, the U.S. Department of Justice announced that Binance would plead guilty and pay $4.3 billion in penalties over widespread AML violations. Binance founder and former CEO Changpeng Zhao also pleaded guilty and agreed to pay a $50 million fine to the U.S. Commodity Futures Trading Commission. The $4.3bn penalty that Binance agreed to pay was not a "goodwill" gesture on the part of Binance or its former CEO or a mitigating sentencing factor which the Sentencing Memorandum appears to imply.[106] Rather it was the reflection of the alleged gravity and widespread behavior by Binance to avoid money laundering regulations over many years. Out of the total penalty, $1.8bn Criminal Fine was discounted 20% of the **bottom** of the applicable Sentencing Guidelines fine range.[107] The $1.61bn was just a disgorgement of fees which Binance collected, according to the settlement with the DOJ, from the unlicensed money transmitting business offense.[108] Lastly, $900 million was the amount of transactions which Binance caused in violation of Iran sanctions.[109]

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[102] The Sentencing Memorandum, p.15.

[103] https://www.chainalysis.com/blog/crypto-laundering/.

[104] *Id*., Chart "Exchanges receiving illicit Bitcoin in 2019."

[105] For a summary of Binance regulatory issues through June 2021, see K2 Integrity report, https://www.k2integrity.com/en/knowledge/policy-alerts/implications-of-mounting-legal-and-regulatory-scrutiny-on-binance/. Also see Reuters Special Report from October 17, 2022, *How Binance CEO and aides plotted to dodge regulators in U.S. and UK*. Also see Reuters Special Report as of January 21, 2022, *Crypto giant Binance kept weak money-laundering checks even as it promised tougher compliance, documents show*. UK's FCA banned Binance Market Limited from undertaking any regulated activity in the UK in June 2021, https://www.fca.org.uk/news/news-stories/consumer-warning-binance-markets-limited-and-binance-group. Monetary Authority of Singapore placed Binance on the Investor Alert List (IAL) in September 2021 for soliciting Singapore users without a license. Earlier, Binance attempted to relocate to Malta in March 2018 after facing regulatory issues in Japan, but failed to obtain the required license and reportedly abandoned its efforts to obtain license in October 2019, which was confirmed by Maltese authorities in February 2020, https://www.mfsa.mt/news-item/public-statement-2020/.

[106] The Sentencing Memorandum, p. 17.

[107] United States of America v. Binance Holdings Ltd., Changpeng Zhao Et al. Memorandum Regarding Plea Proceedings. Para 8.

[108] *Id*., Para 9(i).

[109] *Id*., Para 9(ii).

Notably, on November 21, 2023, Binance also agreed to pay $969 million to settle its potential civil liability for 1,667,153 apparent violations of multiple sanctions programs administered by the US Treasury's Office of Foreign Assets Control (OFAC).[110] As per the language of the OFAC settlement with Binance, "[i]n addition to disregarding known sanctions risks, Binance management also took steps to undermine its own compliance function, encouraging users to circumvent the company's own ostensible controls."[111] Furthermore, OFAC stated that "[t]he settlement amount reflects OFAC's determination that the Apparent Violations were not voluntarily self-disclosed and that Binance's conduct was **egregious**".[112]

As part of the settlement with DOJ and its plea related to money laundering violations, Binance entered into an agreement to undergo a three-year independent monitorship to revamp its compliance and AML policies and procedures. On May 10, 2024, Binance was charged with noncompliance with Canada's anti-money laundering/countering the financing of terrorism (AML/CFT) law. FINTRAC, Canada's national financial intelligence agency, fined Binance $4.4 million over AML/CFT violations which included failure to report large transactions on 5,902 separate occasions and failure to register with FINTRAC or meet deadlines to register.[113] The largest crypto exchange has a long way to go to become compliant.[114]

2. Even Robust KYC/AML Compliance Procedures Do Not Insulate Virtual Asset Service Providers from Exposure to Illicit Finance Transfers

The complaint alleges that Bitzlato received approximately $15mm in crypto transfers from ransomware groups, and the FinCEN Bitzlato Order dated February 1, 2023, apparently seeks to clarify that FinCEN has documented numerous transactions between Conti-associated convertible virtual currency addresses and Bitzlato addresses.[115] Conti is one of the well-known ransomware groups. However, the FinCEN Order is silent as to the source of or any analysis by which the agency identified such alleged numerous transactions. Moreover, while FinCEN Order claimed the existence of significant cryptocurrency transfers, both direct and indirect, between Bitzlato and the Russian sanctioned crypto exchange Chatex, the Raas group DarkSide, and the Phobos ransomware group affiliates, the agency either did not cite any source for such findings, or limited itself to citing June 2021 publications in New York Times and Forbes.[116]

[110] Department of the Treasury, Enforcement Release: November 21, 2023.
[111] *Id*.
[112] *Id*.
[113] https://fintrac-canafe.canada.ca/new-neuf/nr/2024-05-09-eng.
[114] It is remarkable that, despite all the legal actions and compliance violations, Binance has continued to grow aggressively since November of last year. Specifically, Binance's user base has grown by more than 30 million users in the first half of 2024 to reach the 200 million mark. Customer assets have grown from approximately $70bn in late November 2023 to approximately $110bn currently. Also notably, Changpeng Zhao, while no longer Binance's CEO, retains the majority stake in the exchange. See https://www.binance.com/en/square/post/9231338868233, https://defillama.com/cex/binance-cex#information.
[115] FinCEN February 1, 2023, Order "Imposition of Special Measure Prohibiting the Transmittal of Funds Involving Bitzlato", p. 9.
[116] *Id*., footnotes 25 and 28, p. 10.

The blockchain tracing data by Arkham Intelligence which I have reviewed shows no direct transfers of Bitcoin between Conti wallets and Bitzlato wallets. At the same time, Arkham tracing data shows that the US-based Gemini exchange which is one of the first holders of BitLicense from the NYS Department of Financial Services and has been long praised for its robust AML/CFT compliance policies, did have multiple direct transfers of BTC from its wallets to Conti addresses over the span of several years. Specifically, Arkham tracing analysis shows that there was a total of 13 direct transfers of BTC from Gemini hot and other wallets to Conti addresses between May 17, 2019, and May 21, 2021, totaling 762.29 BTC with a contemporaneous aggregate value of $9.9MM USD.[117] Out of this total, 527 BTC were withdrawn from Gemini to Conti ransomware wallets between May and November of 2019.

I am not aware of any criminal enforcement actions against Gemini or its co-founders for what would appear to be a serious “lapse” in its compliance due diligence. But the key conclusion is that it is not commensurate to penalize a virtual asset service provider using the benefit of a perfect hindsight today for its failure to block some past illicit transfers during the time period (in this instance, 2019), when either “bad actor” wallets could have no ownership attribution or at-scale robust blockchain screening tools simply did not exist yet in the crypto industry. This certainly applies in the case of Bitzlato where Mr. Legkodymov and his team, whom I interviewed, until this day are not aware about any details of alleged interactions with ransomware addresses during the period of time when robust on-chain screening tools were not available yet.

IX. The DOJ Today’s Estimates of Bitzlato Exposure to Illicit Finance Are Also Overestimated Compared to What Could Be Known and Prevented at the Time of Such Exposure

The estimate of more than $700 million in direct and indirect crypto funds transfers between Hydra Market and Bitzlato customer wallets alleged in the Amended Affidavit and Complaint and repeated in the Sentencing Memorandum also suffers from the issue of perfect hindsight available to the U.S. Government as of early 2023 (the time of Mr. Legkodymov’s arrest), but surely unavailable to Bitzlato, and to any virtual asset service provider for that matter, at the time of a potentially illicit cryptocurrency

[117] BTC transaction hashes de2e984f86bb53a9013589887b307b9007d9f57f35cd5418d257e012fe8350ba, 9b5217fe34c9f70a28c3b2b9c6e4595c7915c3a9c1583b49b0c7af3b17e6f899, 3f65a73c72daddd5860e371553fea8050574b6ac49f9b13b142b11051d774342, 75fb5a6014d8b0300c63e52d9248397d384f66af88072a19a79c9b888443deb0, 498575397f197b1e9cac6d84211495ee11b2f705ae70a9336456f12947a5110b, 10226a5474e5d603af6b0be7979ebc074b721bd0a5454f872950546f2fbd61d2, fd31c313952168de56945cf9c30d613a81b11d6ee3526ee3077c583db725bbf0, 44709c277cc009d5ceffe41f31d1bd0f69894db38051b03cc9d7f9c12eca0a1a, e36aeb4534234510c8644b21a57bf265d21a9d279b1fcab0b5556e43b0fe3677, 076692790c1fd50d7f40f86c4c2a40ff0606aeeb009157a422004f4d818e87d1, d2a788ad034c783d7a8763386c7c3e6c20246ecae1e0a08fae0db99dbf7b0772, c3d7244b6ea572c8ac4d061fdbf23cdc3fa403ee61ee2de53783be28e59213d3, 8d06caa4a890288105cf3fb9836141b6105ecbcd403eccc585509040eb682f5c. BTC price data is from Cryptocompare.com.

transaction or a real-time wallet screening. The issue of "perfect hindsight" in blockchain analytics arises due to the fact that, unless wallet ownership information (wallet labeling or tagging) becomes available through open-source intelligence information, or information discovery by law enforcement agencies, or through some other off-chain methods, wallet attribution typically requires transactional history for that wallet.

Building transaction history for an address takes time, however. This implies that a peer-to-peer exchange whose customer is receiving a deposit from a newly created address or withdrawing cryptocurrency to a new withdrawal address will not know whom its user is interacting with at the time of such deposit and withdrawal transactions. Only later, once newly created deposit or withdrawal addresses engaged in other transactions as inputs or outputs, may it become possible to attribute them to a particular entity based on the observed transactional patterns and relations with neighboring wallets. If an exchange relies on commercially available wallet and transaction screening software for compliance purposes, the ability of that exchange to detect, block an illicit transfer or initiate a suspicious activity report about such a transfer is further delayed if wallet ownership information is not timely updated in the database of the screening software provider. In this respect, it is instructive to recall that in December 2022, as Bitzlato was switching from Valega to the new blockchain analytics firm Scorechain, Bitzlato shared with the latter its information on crypto addresses for several darknet markets which Scorechain had not been previously aware of.[118]

Wallet labeling also takes time through required research and investigations. One of the first substantive analyses of Hydra Market "ecosystem" using blockchain analytical tools to investigate transactional activity between Hydra wallets and other destinations (mostly centralized crypto exchanges and so-called "risky services") appeared only in May 2021 as a collaboration between Flashpoint and Chainalysis.[119] Furthermore, it took eight months for Germany's Central Office for Combating Cybercrime (ZIT) and the Federal Criminal Police (BKA) in collaboration with the US agencies to investigate and shut down Hydra platform on April 5, 2022.[120] Based on the results of the investigation, the US Treasury's Office of Foreign Assets Control (OFAC) sanctioned and published 117 Bitcoin addresses of Hydra on the same day.[121] And while the majority of Hydra addresses were likely known in early 2021 to mid-2021, a complete picture of the Hydra "ecosystem" and its labeled wallets likely emerged only after the darknet market was shut down in April 2022.

Specifically, in February 2021 Chainalysis identified 4,772,083 addresses as Hydra addresses.[122] However, as of today, based on the investigation by the German and US authorities and OFAC's list of sanctioned Hydra addresses released on April 5, 2022, there are approximately 6.6 million Hydra addresses in total.

Therefore, the DOJ estimate of over $700 million in direct and indirect transfers between Hydra and Bitzlato customer wallets is bound to be too high because it does not remove from the total those transactions which occurred when one or more of the transacting addresses were not known to Bitzlato

[118] See Footnotes 91 and 92.
[119] Flashpoint, Chainalysis, Hydra: *Where The Crypto Money Laundering Trail Goes Dark*, May 2021.
[120] https://www.elliptic.co/blog/5-billion-darknet-market-hydra-seized-by-german-authorities.
[121] https://home.treasury.gov/news/press-releases/jy0701.
[122] https://geminiadvisory.io/alleged-hydra-market-operators-identified-6/.

as a Hydra address due to the absence of ownership attribution. It would only be appropriate to exclude such transactions from the estimated exposure of Bitzlato to Hydra because Bitzlato could not in principle detect such transactions and freeze or ban the associated user account. However, I have not seen any indications or evidence that DOJ has performed such exclusions.

X. Conclusion

In summary, it is my conclusion after reviewing all of the filings and materials provided, that Mr. Legkodymov and Bitzlato did not engage in a conspiracy to transact with darknet Hydra Market or ransomware wallets or to derive any profits from such transactions. Bitzlato was a Russian facing exchange that was specifically trying to avoid the US market and comply with evolving regulations in the nascent marketplace. The exposure occurred due to the fact that Bitzlato was a peer-to-peer trading platform, and solely by virtue of the transactions of its users. Bitzlato's exposure to illicit finance flows, however, was much less than what the U.S. Government has estimated and much smaller than that of much larger cryptocurrency exchanges Binance and BitMEX, whose CEOs were criminally charged in the U.S. on anti-money laundering compliance violations. Furthermore, unlike the executives of Binance and BitMEX, Mr. Legkodymov was taking steps to not willfully violate the Bank Secrecy Act by implementing regulations and pushing for better internal compliance processes. Mr. Legkodymov repeatedly pushed his staff to implement compliance measures to block illicit transactions and clean Bitzlato customer wallets as evidenced by his conduct and the communications with his teams.

Submitted June 21, 2024

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Dr. Boris M Richard, Ph.D.

United States of America v. Anatoly Legkodymov

Criminal Docket No. 23-496

Dr. Boris Richard Expert Report

Supporting Exhibits

Chart 1: Tracing Indirect BTC Flows Between Hydra and Bitzlato

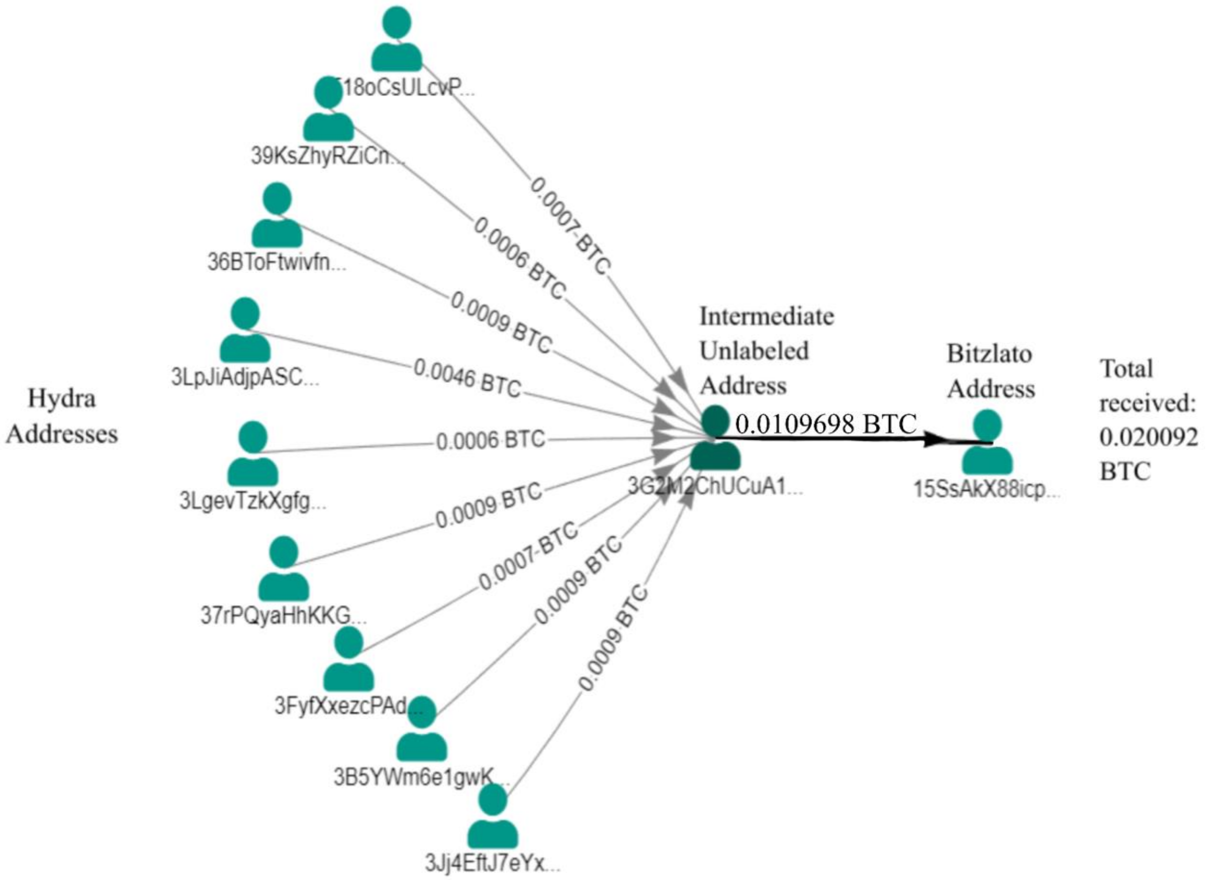
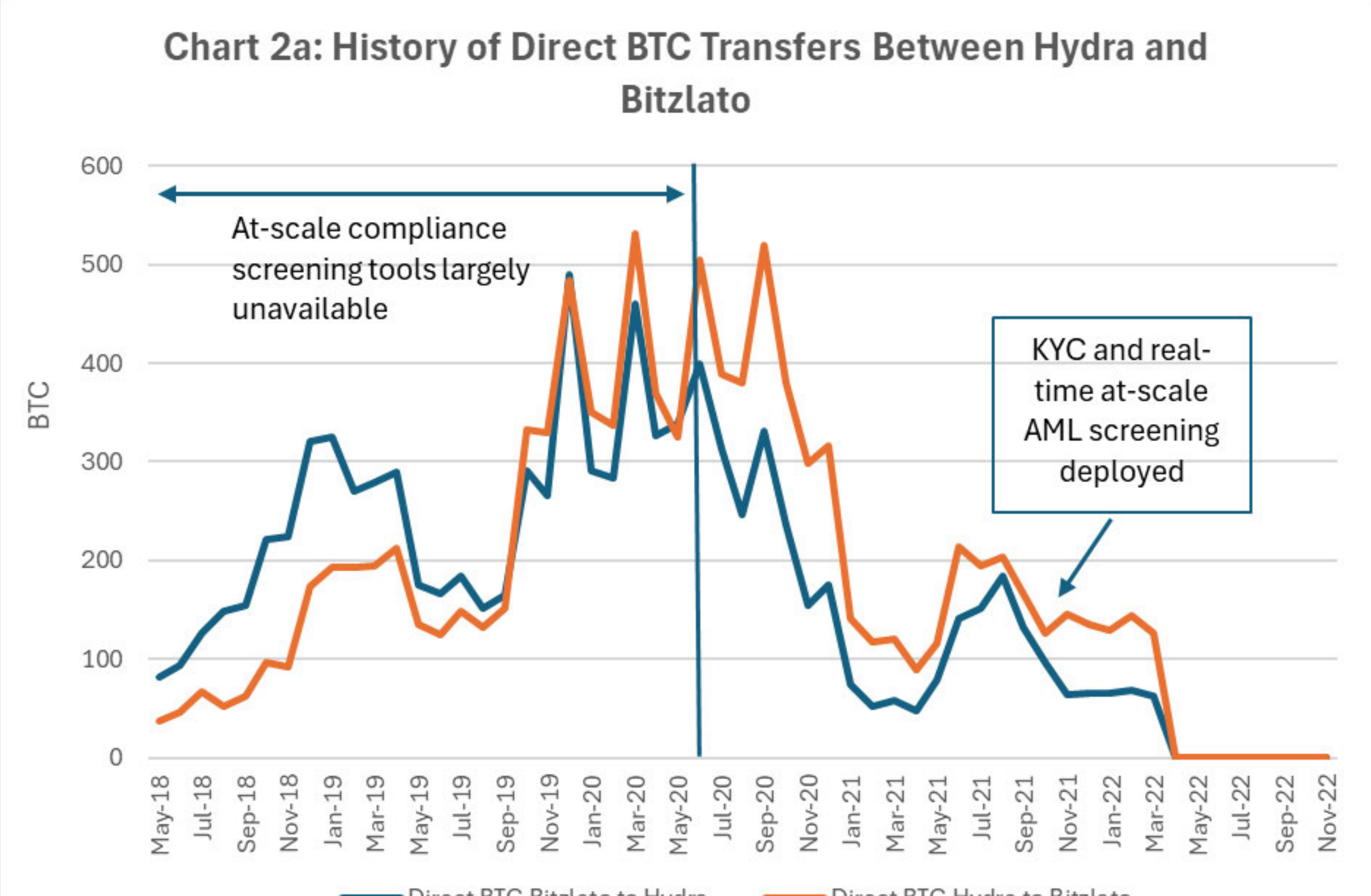


Figure 1: DOJ's Indirect BTC Flows Between Hydra and Bitzlato Are Overestimated by 2.7X-3.7X Times

	Complaint ($USD MM)	Hoptrail ($USD MM) (2 hops)	Crystal ($USD MM) (5 hops)	Complaint (BTC)	Hoptrail (BTC) (2 hops)	Crystal (BTC) (5 hops)
From Hydra to Bitzlato						
Direct Transfers	170.6	171.2	170.3	N/A	10,124.6	10,130.3
Indirect Transfers	218.7	20.5	36.3	N/A	1,380.1	2,390.8
From Bitzlato to Hydra						
Direct Transfers	124.4	121.0	124.9	N/A	8,994.3	9,325.9
Indirect Transfers	191.9	89.0	117.3	N/A	2,937.1	3,614.7
TOTAL	**705.6**	**401.62**	**448.81**		23,436.0	25,461.7
DOJ Total Overestimation (%)		**75.7%**	**57.2%**			
DOJ Indirect Overestimation (%)		**275.3%**	**167.3%**			

Source: Author's analysis, Hoptrail , Crystal.

Chart 2a: History of Direct BTC Transfers Between Hydra and Bitzlato



Source: Author, Crystal.

Chart 2b: History of Indirect BTC Transfers Between Hydra and Bitzlato

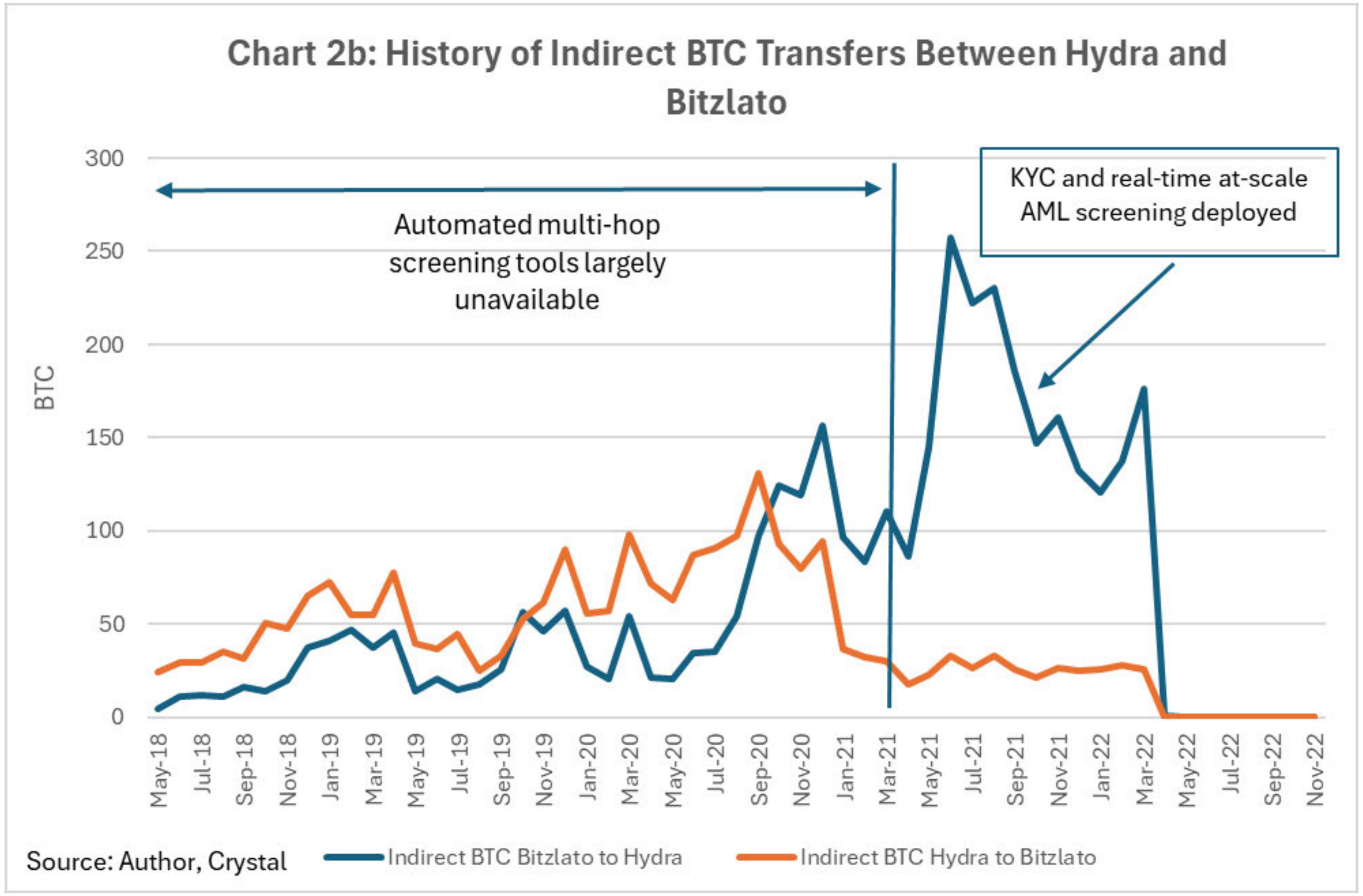


Chart 3: User Account Bans by Bitzlato Technical Support

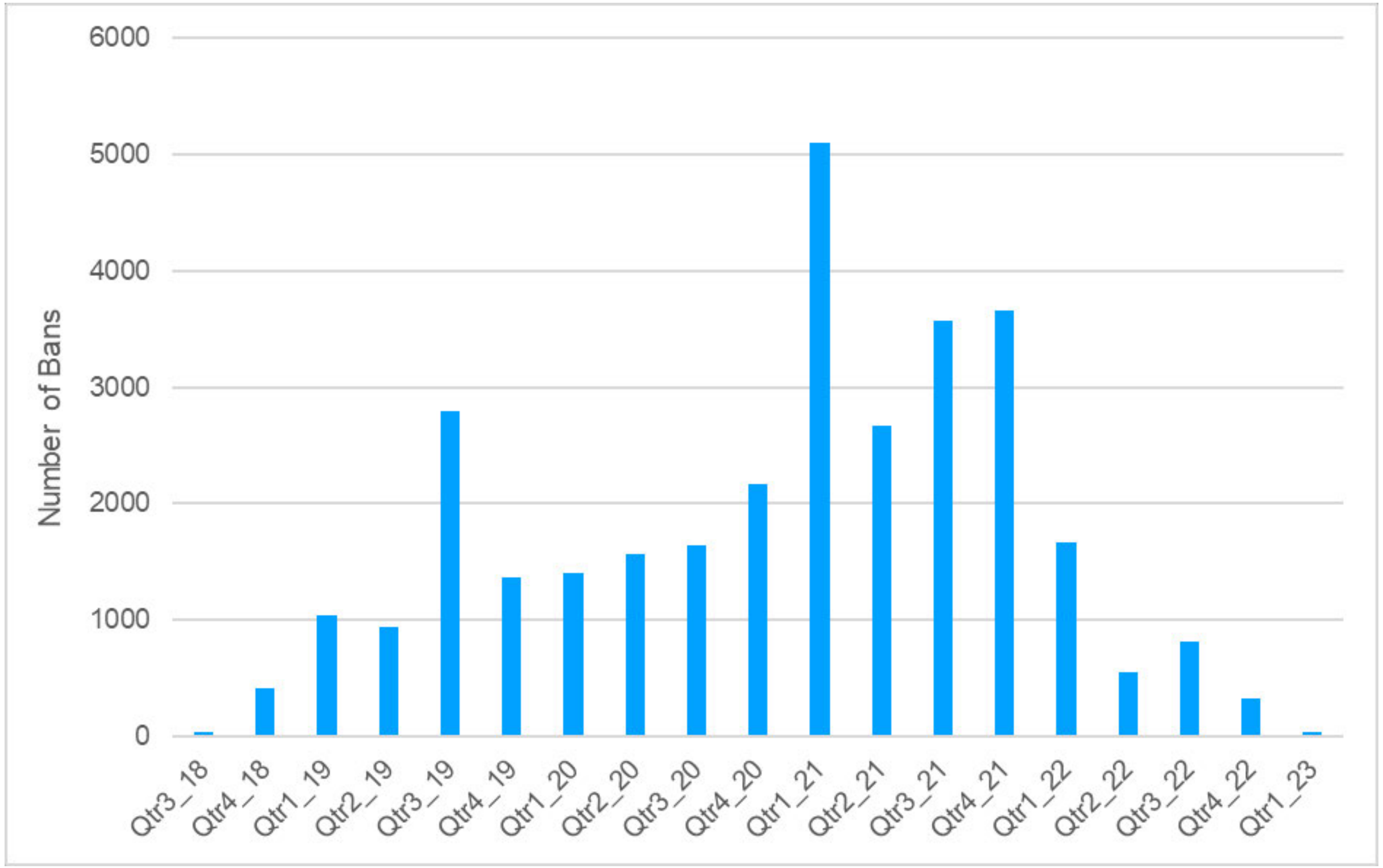


Chart 4: History of Account Freezes by Bitzlato Technical Support Team

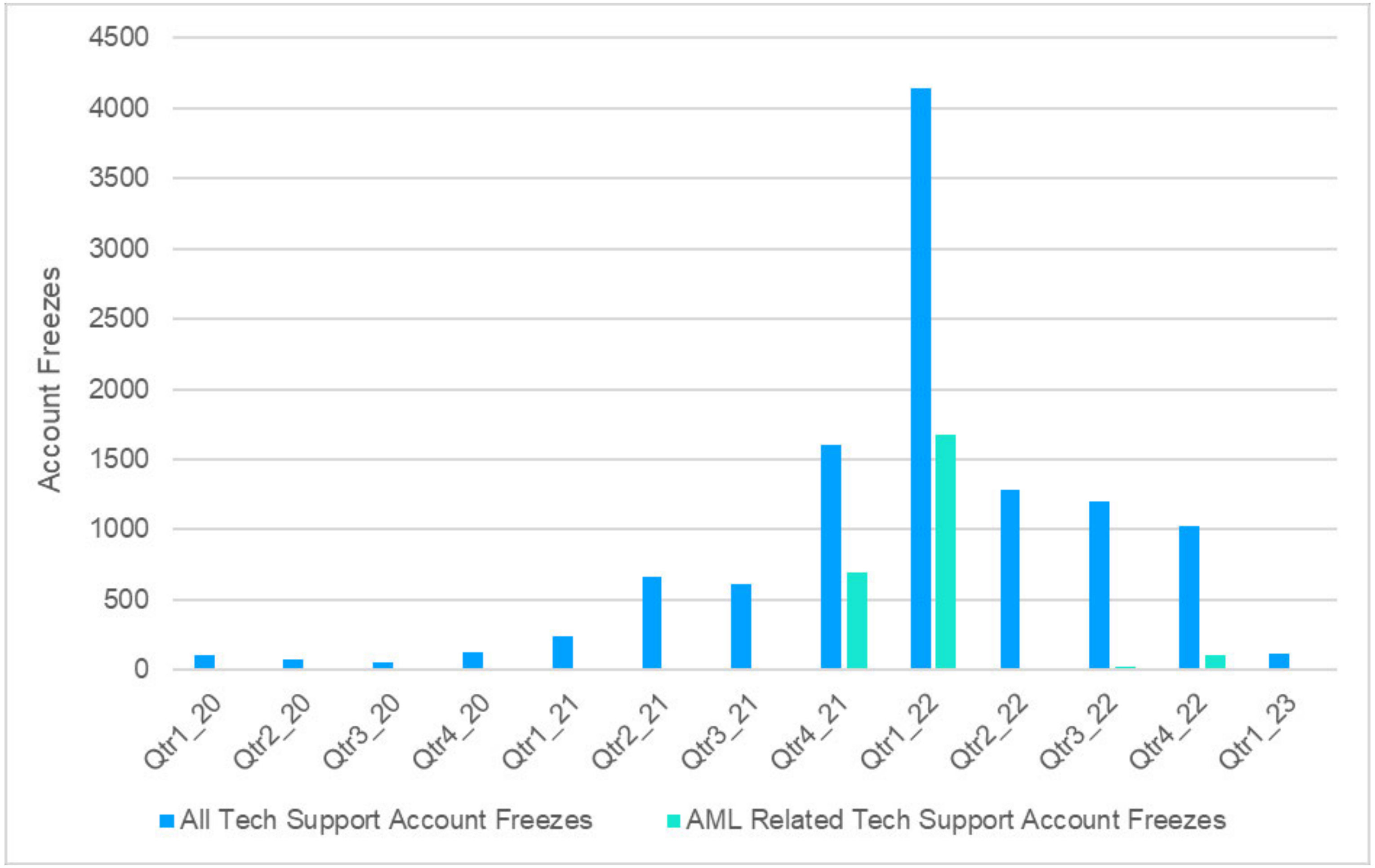


Chart 5: AML Screening-Based Account Freezes by Bitzlato Compliance Team Skyrocketed in 2022

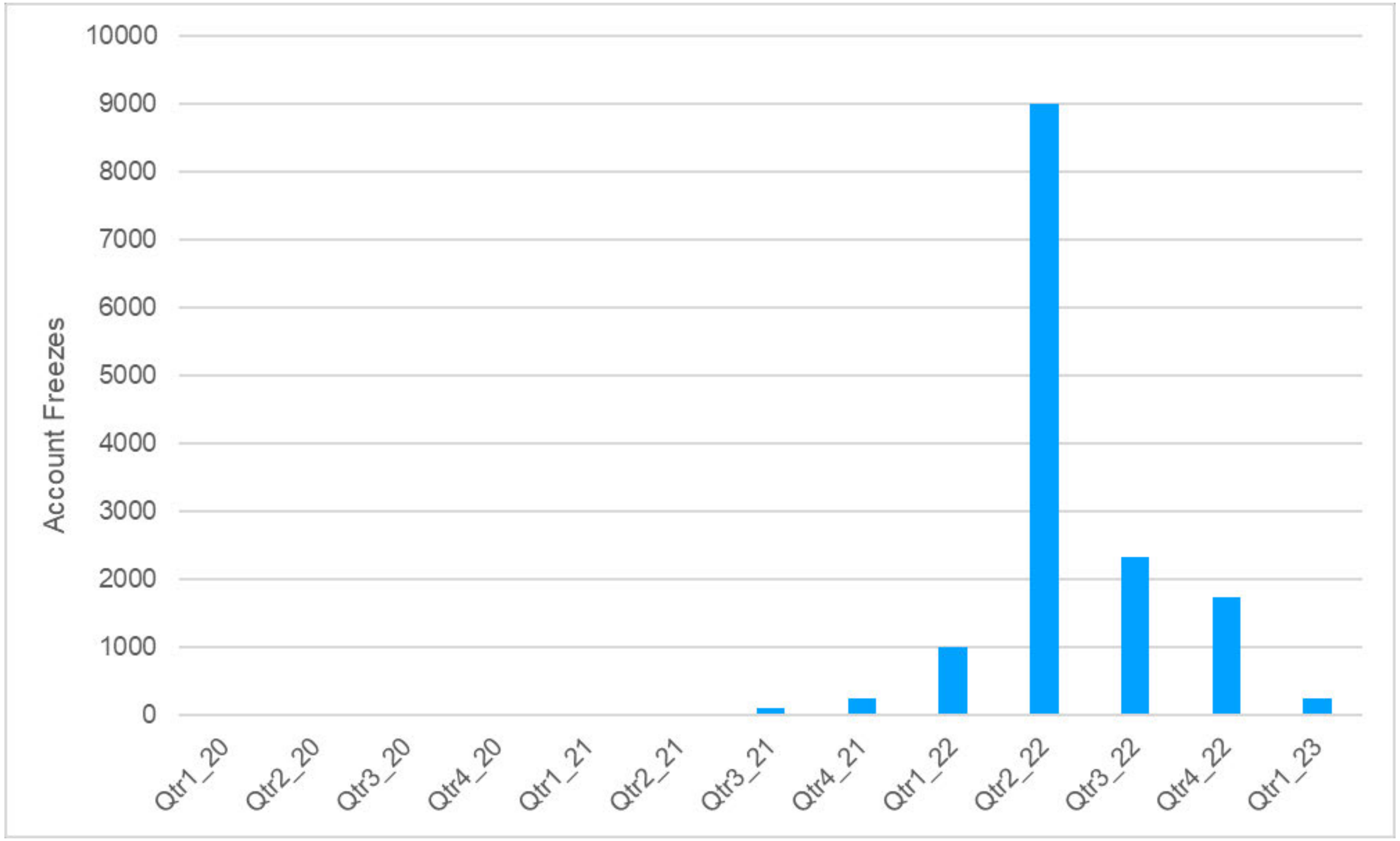


Chart 6: Compliance-Related Account Bans by Bitzlato Compliance Team Ramped Up in 2022 As Well

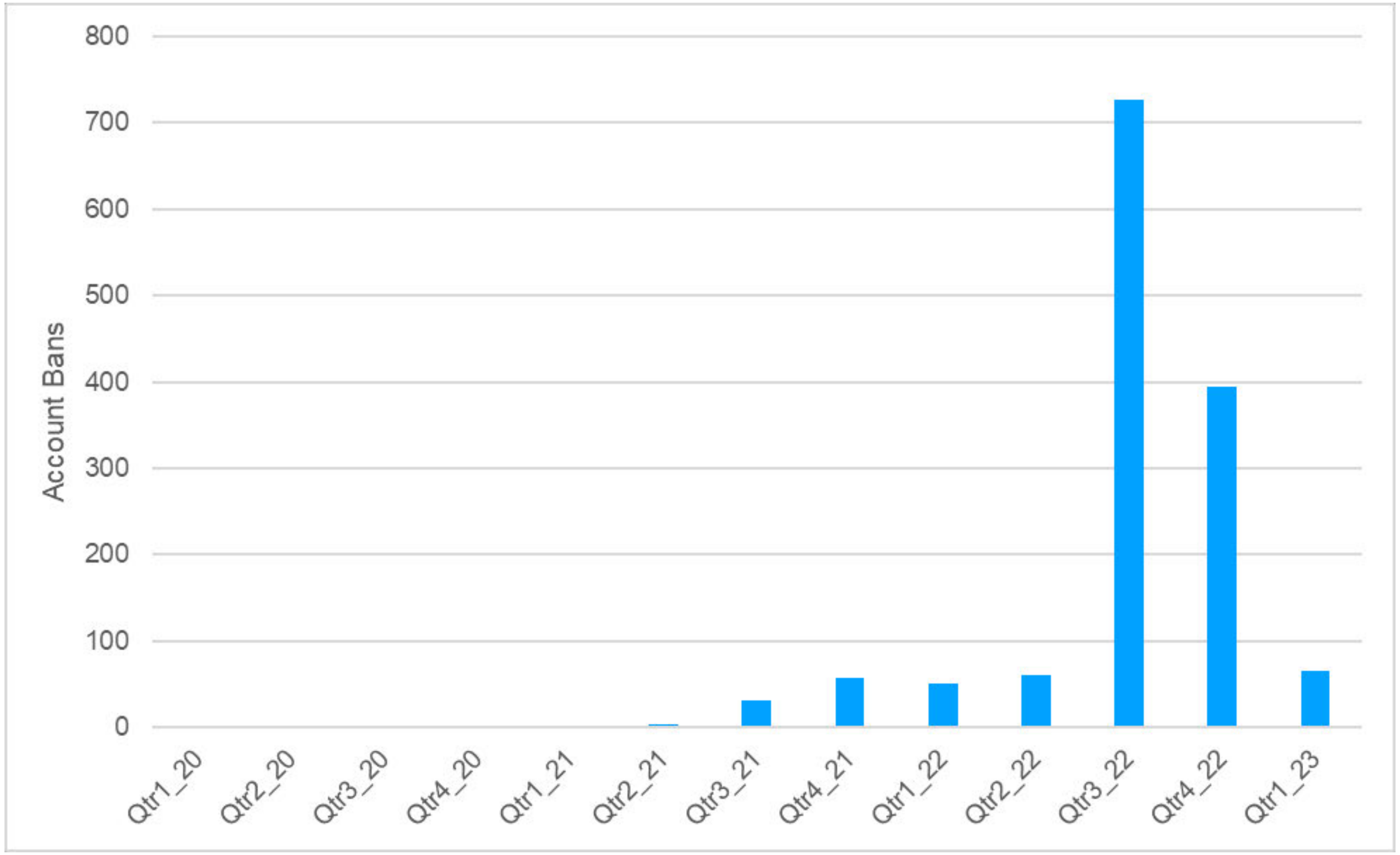


Chart 7: Number of SARs Resulting in Full User Account Blocks

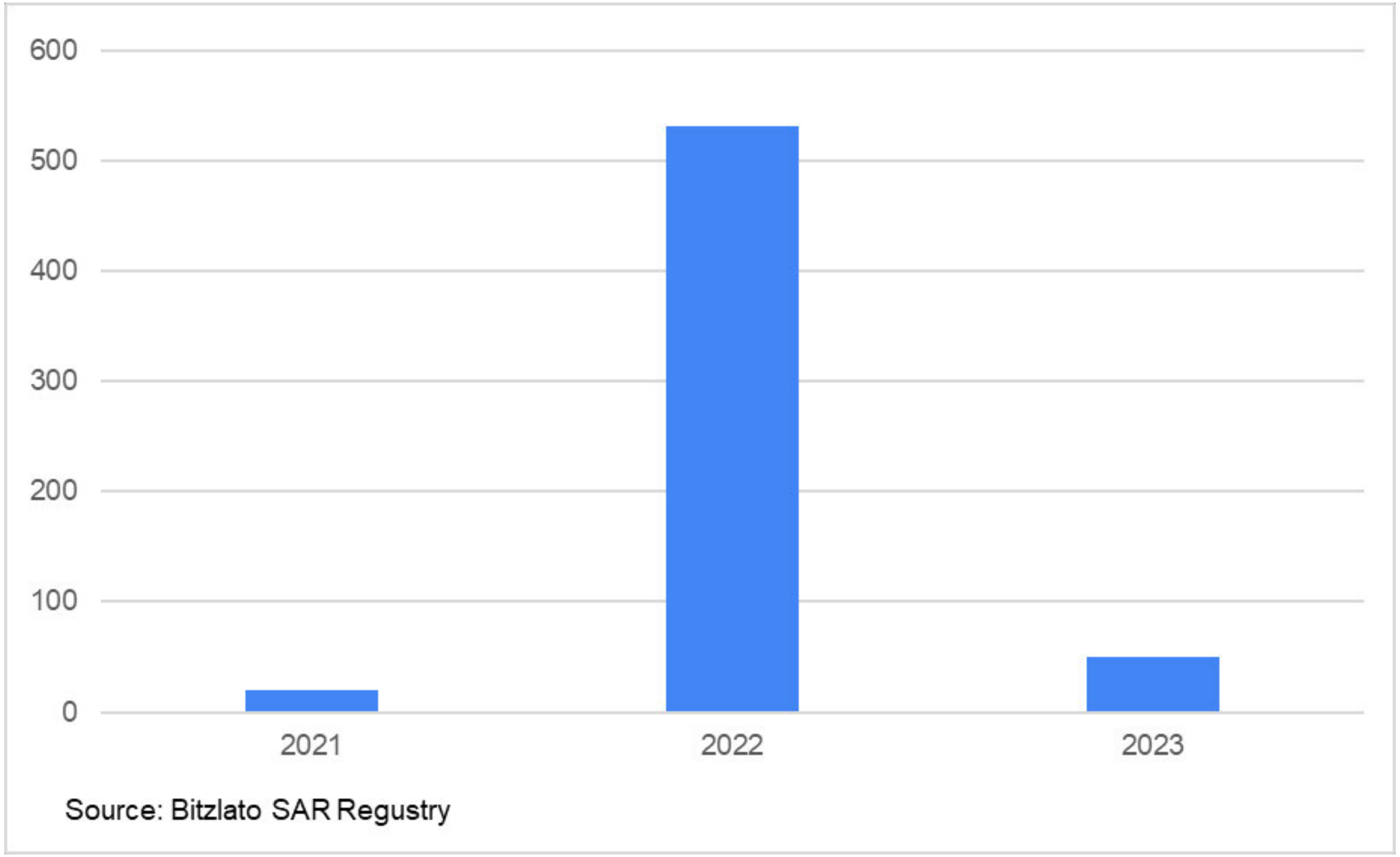


Chart 8: As Bitzlato Compliance Protocols Improved, "Dirty" BTC Deposits Declined to Near Zero in 2022

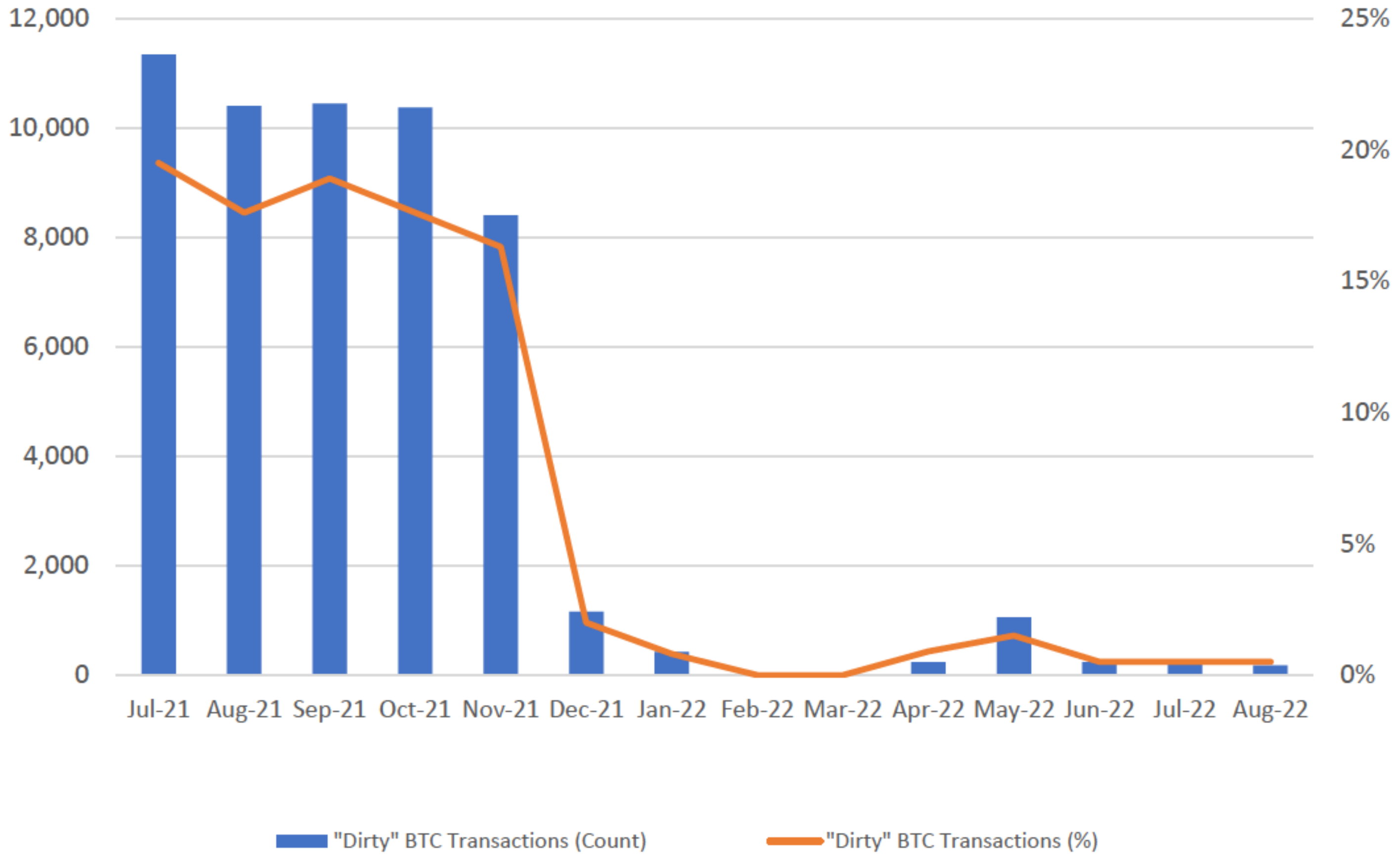


Figure 2: Even Hypothetical Bitzlato's Exposure to US Users Is Miniscule

(Based on Bitzlato's Data on Unverified Users Using U.S. Payment Systems)

User Characteristics (Currency, IP)	By Transaction Volume (%)	By User Count (%)
Only RUB	1.3%	28.4%
Only UAH (Ukranian Hryvnia)	0.1%	10.5%
Both RUB and UAH	0.1%	11.4%
RUB, UAH, other former Soviet Union	0.1%	4.8%
AED (UAE Dirham)	97.5%	27.5%
AED and Russia IP	67.5%	21.0%
Other non-USD currencies	0.0%	1.7%
Users with no info	0.8%	15.7%
Users with no info and non-US IP	0.4%	5.2%
Users with no info and unknown IP	**0.4%**	**10.5%**

Figure 3: Legkodymov's and Bitzlato's Alleged Violations Are Much Lesser Than For Other Criminally Prosecuted Crypto Executives and Their Exchanges

	Binance	BitMEX	Bitzlato
Hydra-Related Transactions (BTC)	80,012	791	25,465
Hydra-Related Transactions ($ millions)	991.4	6.2	448.9
Transactions with Sanctioned Entities	**YES**, $898.6MM with Iran (1, Para 22; 3); $ millions in transactions with Cuba, Syria, Ukraine annexed areas (1, Para 79) ; Russia (4, p.33-34); Terrorist Financing (4, pp. 45-49)	**YES** (8, p.13)	**NO**
Targeting and Solicitation of US Customers	**YES** (1, Para 26-27)	**YES** (7, Para 2, 26; 5, Para 74-75, 78-79; 6, Para 26)	**NO**
Derived Profits from US Customers, Sanctioned Entities, Illicit Finance	**YES**, $1.612bn in profits, including from sanctioned jurisdictions (1, Para 30, 47, 79), (2, Para 4)	**YES**, $1bn in profits (8, p. 30)	**NO**
Substantial Presence in the United States	**YES**, 20-30% of business and 1/3 of customers in 2018 (1, Para 28-29, 36), more than $3.4bn in tx with Western District of Washington (1, Para 83b), (2, Para 77, 95), $35bn in US tx in a month (4, p.8) , 2.51 mm US customers (2, Para 138)	**YES**, 4-8% are US customers (7, Para 17; 8 , pp. 3, 12-13), $11bn in BTC deposits, 85,000 US users (5, Para 3)	**NO**
Control Persons Knew About Substantial Illicit Finance Exposure and KYC/AML Violations	**YES** (1, Para 54, 58, 64-66, 71; 4, p. 32; 2, Para 95)	**YES** (7, Para 24-25, 26; 8, p.13-14; 5, Para 88-92, 99, 136)	**YES** (Illicit Finance)/**N/A** (KYC/AML violations)
Control Person(s) Deliberately Avoided Implementing KYC/AML Policies and Procedures	**YES** (1, Para 23, 33, 58, 71), Willful violation of BSA and implementing regulations (4, p. 9-11, 53)	**YES** (7 (Para 24-25, 28); Willful Violation of BSA, Wilfully Failed to Implement AML Program 8 (pp. 5-7, 14)); (5, Para 88-92, 99, 136)	**NO**

Sources: Crystal, (1) United States of America v. Binance Holdings Ltd, (2) CFTC v Changpeng Zhao, Binance Holdings Ltd et al., (3) United States of America v. Binance Holdings Ltd., Changpeng Zhao Et al. Memorandum Regarding Plea Proceeedings, (4) FinCEN Consent Order in Re: Binance Holdings Ltd, (5) CFTC v HDR Global Trading Ltd et al., (6) CFTC v HDR Global Trading Ltd et al Consent Order, (7) United States of America v Arthur Hayes et al., (8) FinCEN Assessment of Civil Money Penalty in Re: HDR Global Trading Ltd.

Appendix 1
Dr. Boris M. Richard, PhD

Confidential

Key Expertise

- Digital assets and cryptocurrency markets
- Blockchain investigations
- NFT and Web3 economics
- Asset valuation
- Market structure and market manipulation
- Investment and trading
- Fixed income securities and derivatives
- Mortgage-backed securities
- Structured cash flows

Education

Ph.D., Economics, Krannert School of Management, Purdue University, IN, 1996

Project Geographical Experience

U.S., Canada, Europe, Singapore, Japan, South Korea, Brazil

Languages

English
Portuguese

Summary of Experience

Dr. Boris Richard is the Founder of UpAI, LLC, a company providing crypto advisory, and crypto and financial markets litigation expert services. He has over 25 years of experience in institutional fixed income and derivatives markets, as well as consulting and expert work in complex litigations, arbitrations, and investigations. Dr. Richard focuses on cryptocurrencies, blockchain economics and token valuation, token offerings, as well as cryptocurrency related expert work, virtual asset tracing investigations, and responses to regulatory inquiries and enforcement actions. His clients include major law firms, cryptocurrency exchanges, digital asset issuers, custodians, Decentralized Finance platforms, blockchain foundations, and institutional investors.

Prior to his work in digital assets, Dr. Richard had spent 15 years in mortgage-backed securities, fixed income and derivatives markets as market strategist, bond trader and quantitative portfolio manager at Goldman Sachs, Barclays Capital, HSBC Securities, and several large multi-strategy and macro hedge funds.

Speaking Engagements

Dr. Richard presented at various public forums and chaired multiple discussion panels of digital asset industry executives and attorneys on a variety of cryptocurrency related issues, including regulatory and litigation issues, digital asset compliance, cryptoasset valuations, risk and structure of cryptocurrency markets. He is a repeated presenter at Horasis, Knowledge Group webinars, and CPA Academy Webinars.

Professional Affiliations/Memberships/Licenses/Training

Wall Street Blockchain Alliance, Senior Fellow at the International Congress of Blockchain Legal Advisors

Current Focus

Dr. Richard serves as a testifying and non-testifying expert in blockchain and cryptocurrency related litigations and assists clients in issues related to digital asset investigations, crypto asset tracing and recovery, suspected market manipulation, as well as regulatory and enforcement actions. Additionally, he provides advisory services involving cryptoasset valuation, regulatory and business due diligence of crypto native companies, and digital asset industry analysis, including NFTs and Web3 services.

Contact

+1 917-741-5041 (M) | richard@1upai.ai | LinkedIn: linkedin.com/in/boris-richard-2a158ab

Work Experience

UpAI, LLC, Founder, 2024 - present
J.S. Held, LLC, Senior Managing Director, 2022 - 2024
FTI Consulting, Managing Director (2018 - 2022)
Cornerstone Research, Senior Economist and Senior Manager (2011-2018)
Graham Capital Management, Portfolio Manager (2010 – 2011)
Pine River Capital Management, Portfolio Manager, 2010
Platinum Grove Asset Management, Portfolio Manager/Strategist (2006-2008)
HSBC Securities (USA) Inc., Senior Vice President (2003 – 2006)
Barclays Capital, Director (2002 – 2003)
Goldman Sachs, Vice President (1998 – 2002)
Prudential Securities, Inc., Associate Vice President (1996 – 1998)

Select Project Experience

Cryptocurrency and Blockchain Advisory Projects, Investigations, Arbitrations and Litigations

Submitted an arbitration expert report analyzing and opining on wash trading and other manipulative trading practices in certain stablecoins on a decentralized crypto exchange operating on a major public blockchain. Established common ownership of thousands of blockchain addresses through clustering analysis.

Retained as testifying expert in several lawsuits brought by a regulatory agency against token issuers, alleging security registration violations and market manipulation in native cryptocurrency, as well as fraud.

Retained as testifying expert in a lawsuit brought by a regulatory agency against a major token issuer, alleging security registration violations, submitted report and deposition.

Provided expert support involving economic analysis and security review of blockchain ecosystem, its code base, examination of available decentralized applications and consensus protocol in litigation brought by a US regulator against major ICO issuer.

Performed expert work to establish economic uses of a digital token and its value sources, to analyze the degree of decentralization of token holdings and the distribution of token purchases among investors in litigation brought by a US regulator against a non-US ICO issuer.

On behalf of a token issuer responding to a regulatory inquiry, performed expert work involving tracing of cryptocurrency investment funds, verification of reported ICO data and public disclosures by the issuer against blockchain address and transaction data, ownership attribution of various Bitcoin wallets and constituent addresses, as well as analysis of token safe smart contract code to explain certain token transfers.

Engaged by a major financial institution to evaluate AML/KYC/Financial Crime Compliance policies and procedures of certain Virtual Asset Service Providers as part of onboarding process.

Provided expert support on behalf of a major token issuer in relation to investors' arbitration claims involving blockchain platform development.

Retained as consulting expert to prepare a market sounding report for a NFT/Web3 business initiative by an alternative investment manager.

Retained as valuation expert by a blockchain start-up company to prepare a valuation report of its cryptoasset holdings for the purposes of corporate rollover.

Retained as testifying damages and valuation expert to evaluate Proof-of-Stake cryptoasset damages in a dispute between blockchain company co-founder and blockchain developer/token issuer.

Retained as testifying expert to value a large cryptocurrency estate as part of a dispute with the IRS.

Retained as a testifying expert by a cryptocurrency mining company to perform blockchain tracing of several crypto assets and to assess damages resulting from the alleged theft of said assets.

Retained as cryptocurrency expert in several litigations (including marital disputes) and associated blockchain and wallet investigations to trace and recover hidden crypto assets.

Retained as an expert to analyze cryptocurrency transactions/trading activities and to publish report related to Terraform Labs and Luna Foundation Guard and the May 2022 depegging of Terra USD stablecoin.

Retained as testifying expert to opine on the depegging of stablecoins USDC and DAI in the wake of the Silicon Valley Bank collapse.

Insider Trading and Anti-Trust Investigations into Broker-Dealer Trading Practices

Retained as consulting expert by a major broker-dealer to analyze alleged insider trading activities in Treasury bonds, interest rate swaps, forward rate agreements and OIS.

Retained as consulting expert by a major international bank to analyze and opine on trading communications and related trading activities in government bonds, Supra, Sovereign and Agency (SSA) bonds, interest rate derivatives (interest rate swaps, futures, swaptions and forward rate agreements), FX forwards, swaps and options, and certain equity index derivative instruments.

Retained by a primary dealer to analyze and opine on communications and trading activities related to When-Issued (WI) US Treasuries and Treasury auctions.

Expert work to analyze and opine on communications and specific transactions related to regulatory inquiries with respect to US agency debt syndications.

Fixed Income Securities and Derivative Instruments Valuation

Performed expert support work to value REIT preferred stock with embedded options, conduct damages analysis, and rebut opposing expert report.

Provided consulting expert support in municipal ARS preferred cases and VRDO arbitrations.

Retained as consulting expert to analyze performance of trades involving interest rate swaps, Treasuries and LIBOR forward rate agreements.

Retained as consulting expert to assess investment returns of a large portfolio of emerging market bonds as part of a DOJ criminal probe.

Residential Mortgage-Backed Securities

Expert support work in multiple RMBS loss causation matters involving statistical estimation of the impact of various factors on deal and loan performance, rebuttal of the opposing experts' arguments and econometric studies.

Expert support work in multiple RMBS due diligence cases involving a review of underwriters' due diligence processes, as well as loan-level due diligence review of specific acquisition collateral pools.

Engaged in multiple rebuttals of opposing damages calculations; performed affirmative RMBS Section 11, Section 12 and Blue Sky damages calculations.

Expert support work in several cases to rebut opposing expert calculations of damages in connection with alleged breaches of loan representations and warranties, performed affirmative damages calculations.

Performed econometric modeling of deal- and loan-level defaults and RMBS losses; analyzed the default and loss performance of at-issue loan groups and individual RMBS certificates.

Retained as non-testifying expert to conduct valuation analyses with respect to RMBS derivatives backed by reverse mortgage loans.

CDO/CDS and Credit Derivatives

Expert support work to analyze credit exposure in synthetic ABS CDOs backed by subprime mortgage collateral, and to write expert report in response to a regulatory enforcement action against a major CDO issuer.

Engaged in litigation expert analysis related to corporate synthetic bespoke CDO and performance of the CDO collateral manager.

Performed synthetic CDO modeling and analysis of trading activities in CDO tranches and single-name corporate and ABS CDS.

Expert support work to analyze CDO collateral acquisition practices and the role of asset-backed CDS indices (ABX) in structuring CDOs, and to perform economic and valuation analysis to assess potential conflicts of interest between CDO arranger and investors.

Public Engagements

ABA Financial Crimes Enforcement Conference Podcast, "Current State of the Cryptomarket and Major Risks Facing Cryptocurrencies", November 2018.

Organized and conducted several Webinars and CLE events in partnership with major law firms on blockchain and crypto currency regulatory and enforcement issues.

- FTI Consulting All-Day Litigation Forum, blockchain/cryptocurrency panel together with Baker Hostetler, New York, Yale Club, October 2018.
- Webinar with Akin Gump LLP, "What Should Crypto Funds Expect From the SEC Going Forward?", December 2018.
- Webinar with law firm Bernacchi Chambers, "The Potential Impact of Cryptoassets in the Matrimonial Law World", June 2021.
- CEFPRO's Fraud and Financial Crime USA Virtual Annual Event, "Understanding Risks with Cryptocurrencies", October 2021.

Organized panel on cryptocurrency litigation and regulation at the AICPA Forensic & Valuation Services Conference, Las Vegas, November 2019.

Repeated presenter at the CPA Academy Webinars on issues of blockchain and digital assets.

- Webinar "A Look into the Future of Blockchain Litigation", March 2020.
- CPE Webinar "Current Trends in Digital Asset Litigations and Investigations", August 2020.

Repeated presenter at the Knowledge Group Webinars on issues of blockchain and digital assets.

- Webinar "SEC's Heightened Enforcement on Cryptocurrency: What You Should Know and Do to Avoid Legal Risks", June 2020.
- CLE/CPE Webinar "The Rise of Cryptocurrency: What the Future Holds", February 2021.

The Securities Enforcement Forum West 2020, speaker at the panel "Silicon Valley Spotlight – Regulation, Enforcement and Prosecution Trends", May 2020.

Speaker at the Wall Street Blockchain Alliance Cryptos Investors Summit, November 2020.

Host of several discussion panels

- CalCPA Fraud and Forensic Accounting Virtual Conference, discussion panel "Follow the Money, Virtual and Otherwise: Bitcoin and Cryptocurrency. Investigating Cryptocurrency Trails and Tracing Digital Assets", April 2021.
- Horasis Extraordinary Event on the USA, chaired digital asset industry panel "Creating Trust in Digital Asset Markets", March 2021.
- Horasis Global Meeting, chaired digital asset industry panel "Digital Asset Compliance: Why Can't We All Agree?", June 2021
- Horasis USA Meeting, chaired digital asset industry panel "", March 3, 2022

Presenter at Solidus Labs Webinar, "The Dark Side of Token Design – Challenges of Market Integrity and Market Manipulation in DeFi", July 2022.

URMIA panel discussion, "The Emerging Risk of Cryptocurrency in Higher Education, September 8, 2022.

Presenter at the LA Blockchain Summit 2022 industry panel "What's up with the Markets?", November 1, 2022.

Presenter at the Wall Street Blockchain Alliance webinar "Binance, FTX and the Future of Crypto Exchanges", November 11, 2022.

Selected Publications and Materials

"What the Howey Test Misses About Crypto Assets", Law360, June 2019.

"Financial Crimes Compliance for Cryptocurrency: Why Can't We All Agree?", FTI Consulting, April 2021.

Luna Foundation Guard – Technical Audit Report, November 2022.

CLE video "On Chain Analytics In Crypto Litigation", The International Congress of Blockchain Legal Advisors, 2023

Expert Testimonies

SEC v. Token Issuer, April 2022.

Blockchain co-founder v. blockchain developer and token issuer, September 2022.

EXHIBIT B

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EXHIBIT C

id	user_id	created_at	ends_at	freeze_type	reason	active	admin_code
2607753	14684378	06.05.2022 21:59	06.05.2122 21:59	all	Грязный вывод #28(	t	system